

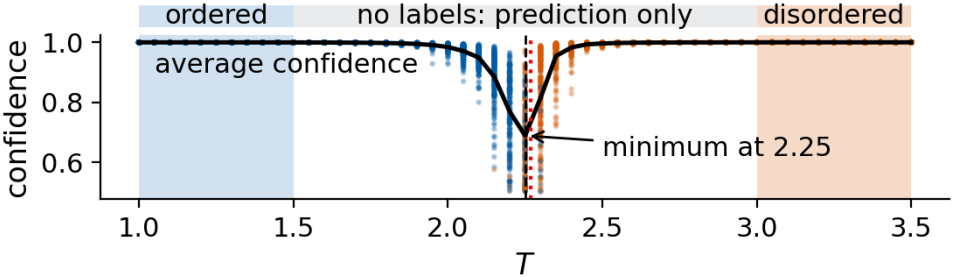
Lecture 5: Unsupervised Learning
PCA, t-SNE, Clustering, Autoencoders
Phase Diagrams of Classical Spin Models without Labels

Deep Learning in Classical and Quantum Physics

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Finding T_c with Labels



- a convolutional neural network (CNN) on 50×50 Ising configurations, labelled "ordered" ($T < 1.5$) and "disordered" ($T > 3$)
- lowest mean confidence at $T = 2.25$ (step 0.05); exact $T_c = 2.269$
- the labels came from us: two phases, and where each lies

Today: the same kind of data, without labels.

1

Learning without Labels



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Supervised and Unsupervised Learning

Supervised	Unsupervised
data: pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$	data: samples only, $\{\mathbf{x}_i\}_{i=1}^n$
goal: $f(\mathbf{x}) \approx y$, i.e. $p(y \mid \mathbf{x})$	goal: structure of $p(\mathbf{x})$
the question is fixed in advance	no question fixed in advance

Labels are expensive, and they contain what we already know.

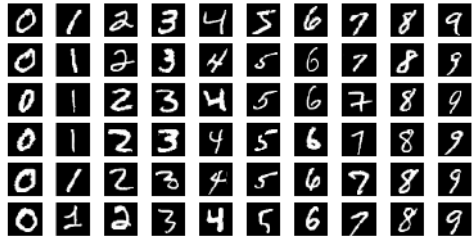
Unsupervised Methods Answer Different Questions

what we learn about the data	method
a few variables per sample	dimensionality reduction
the kinds of samples	clustering
what is typical, what is unusual	anomaly detection
the distribution, new samples from it	generative models

The method proposes a structure; its meaning is for us to decide.

Data Fill a Thin Part of a Large Space

- a sample is a vector $\mathbf{x} \in \mathbb{R}^D$:
  $D = 784$ pixels, or $D = L^2$ spins
- random pixels give noise, never a digit
- dimensionality reduction:
  $\mathbf{x} \in \mathbb{R}^D \mapsto \mathbf{z} \in \mathbb{R}^d$, $d \ll D$



MNIST handwritten digits, 28×28 pixels

2

Principal Component Analysis

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Principal Component Analysis Finds the Directions of Largest Variance

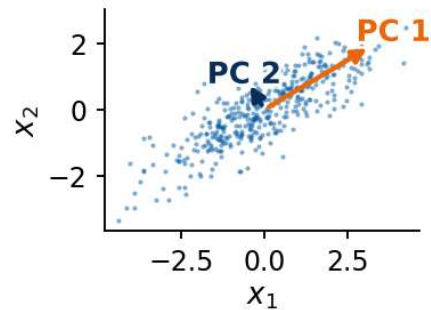
covariance of the features, $D \times D$:

$$C_{jl} = \frac{1}{n} \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l)$$

x_{ij}: feature j of sample i; $\bar{\mathbf{x}}$: mean sample

$$C\,\boldsymbol{\lambda}_l = \sigma_l^2\,\boldsymbol{\lambda}_l, \qquad \sigma_1^2 \geq \sigma_2^2 \geq \ldots$$

$\boldsymbol{\lambda}_l$: principal direction; σ_l^2: variance along it



400 points in $\mathbb{R}^2$; PC 1 (first principal direction) carries 93.4% of the variance

K. Pearson, Phil. Mag. **2**, 559 (1901), doi:10.1080/14786440109462720; H. Hotelling, J. Educ. Psychol. **24**, 417 (1933), doi:10.1037/h0071325

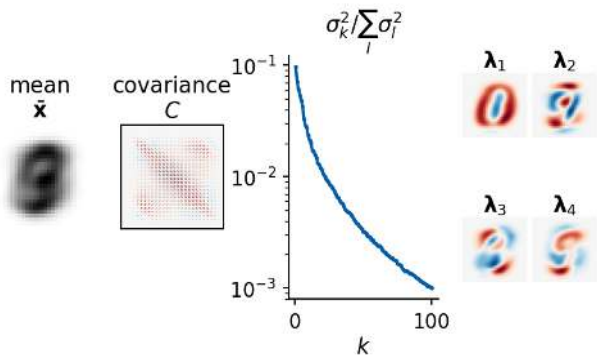
Each Sample Becomes a Few Coordinates

coordinate of the centred sample along $\boldsymbol{\lambda}_l$:

$$w_i^{(l)} = \boldsymbol{\lambda}_l^\top (\mathbf{x}_i - \bar{\mathbf{x}}), \qquad \mathbf{w}_i = \Lambda_k^\top (\mathbf{x}_i - \bar{\mathbf{x}}) \in \mathbb{R}^k, \qquad \Lambda_k = (\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_k)$$

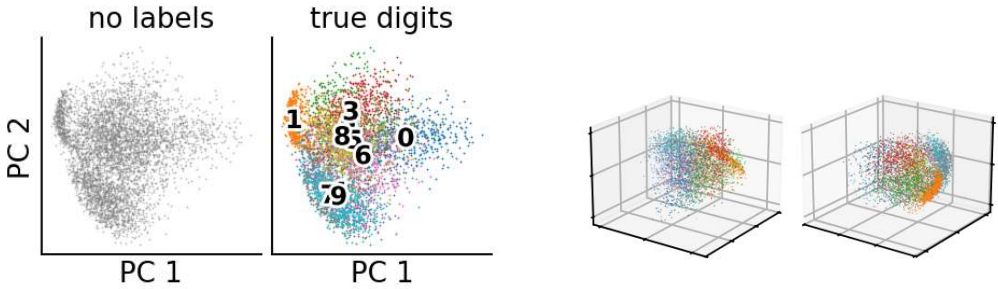
- kept variance: $\sum_{l \le k} \sigma_l^2 \Big/ \sum_l \sigma_l^2$
- $k = 2$: $\mathbf{w}_i$ is a point in the plane
- a fixed linear map: it applies to new samples

PCA of the Handwritten Digits



- mean image $\bar{\mathbf{x}}$, covariance C (784×784), variance fraction $\sigma_k^2 / \sum_l \sigma_l^2$, directions $\boldsymbol{\lambda}_l$ (red > 0, blue < 0)
- 2 components keep 17% of the variance, 10 keep 49%, 87 keep 90%
- $\boldsymbol{\lambda}_l$: pixel patterns that vary together

A Linear Projection Mixes Most Digits

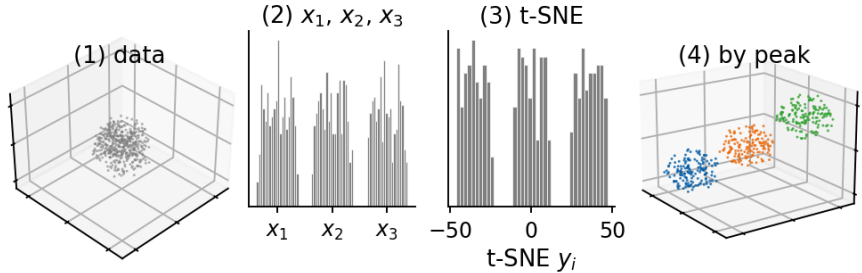


- 5000 test digits on PC 1, 2 (17% of the variance) and on PC 1–3 (23%)
- 0s and 1s separate; most digits overlap

3
t-SNE

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t-SNE Finds Three Groups that No Coordinate Shows



- 450 unlabelled points in $\mathbb{R}^3$: one blob (1); every coordinate overlaps (2)
- t-SNE (t-distributed stochastic neighbour embedding) gives each point one number y_i: three peaks (3)
- colouring by peak reveals the three generating groups (4)

Neighbour Probabilities in the Data

$$p_{j|i} = \frac{e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2}}{\sum_{l \neq i} e^{-\|\mathbf{x}_i - \mathbf{x}_l\|^2/2\sigma_i^2}}, \qquad p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$$

- $p_{j|i}$: probability that $\mathbf{x}_i$ picks $\mathbf{x}_j$ as its neighbour
- width σ_i set by the **perplexity**, roughly the number of neighbours (here 30)
- $P = \{p_{ij}\}$: a distribution over pairs, $\sum_{i \neq j} p_{ij} = 1$

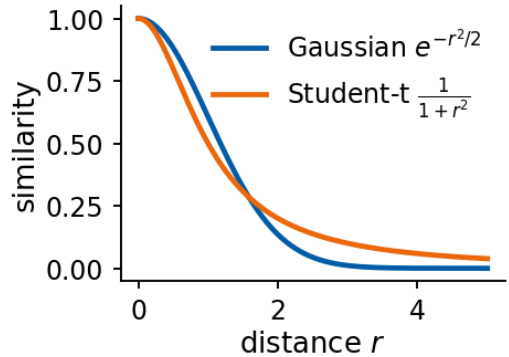
Neighbour Probabilities in the Map

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{l \neq m}(1 + \|\mathbf{y}_l - \mathbf{y}_m\|^2)^{-1}}$$

- $\mathbf{y}_i \in \mathbb{R}^d$: the map point of sample i; $d = 2$ to plot
- the $\mathbf{y}_i$ are the unknowns, free parameters
- $Q = \{q_{ij}\}$: the same kind of distribution, in the map

The Heavy Tail Opens Gaps between Groups

- 2D has no room for many neighbours at similar distances (*crowding problem*)
- the Student-t decays slowly: moderately distant samples move apart
- gaps open between groups



similarity against distance r: Gaussian $e^{-r^2/2}$ (data), Student-t $1/(1+r^2)$ (map)

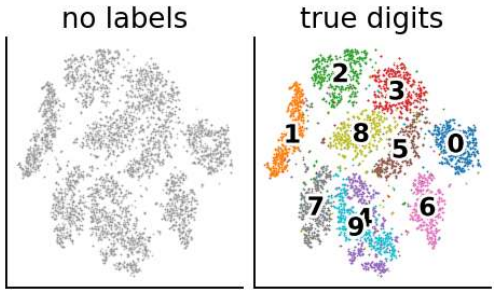
t-SNE Matches the Two Distributions

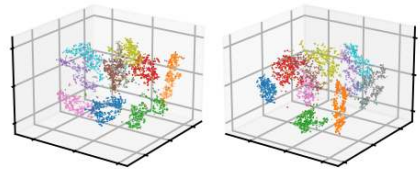
$$\mathcal{L}(\mathbf{y}_1, \ldots, \mathbf{y}_n) = \mathrm{KL}(P \,\|\, Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

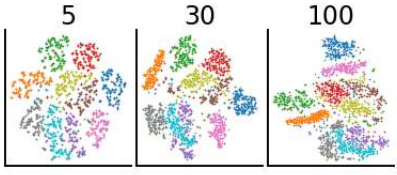
- Kullback–Leibler (KL) divergence, minimised by gradient descent in the $\mathbf{y}_i$
- neighbours placed far apart cost much: the **local** structure is kept
- no formula $\mathbf{x} \mapsto \mathbf{y}$: no map for new samples

L. van der Maaten, G. Hinton, J. Mach. Learn. Res. **9**, 2579 (2008), jmlr.org/papers/v9/vandermaaten08a

t-SNE Separates the Digits



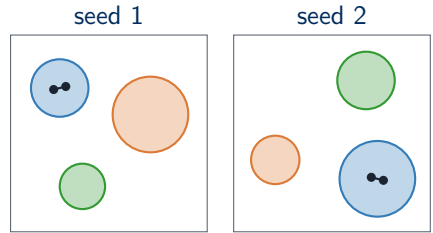




top: 5000 digits in 2D and 3D, labels only colour the points; left: 2000 digits at perplexity 5, 30, 100

Only the Neighbourhoods in a t-SNE Map Are Reliable

- **not meaningful:** positions of groups (another seed moves them)
- **not meaningful:** gaps and sizes of groups
- **reliable:** which samples are close neighbours



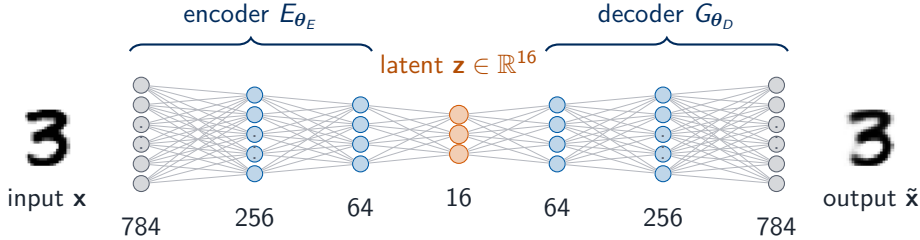
Schematic: two runs on the same data. Groups move, mirror and resize; two neighbours (black) stay together.

4

Autoencoders

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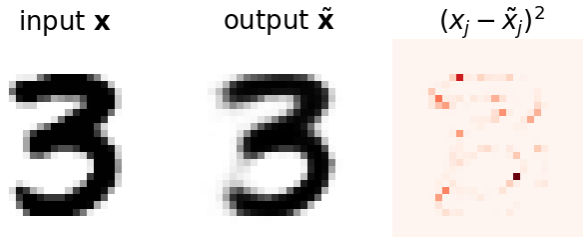
Autoencoder



- encoder $\mathbf{z} = E_{\theta_E}(\mathbf{x}) \in \mathbb{R}^d$, decoder $\tilde{\mathbf{x}} = G_{\theta_D}(\mathbf{z}) \in \mathbb{R}^D$
- the bottleneck $d \ll D$ keeps what the decoder needs: a **latent representation**

G. E. Hinton, R. R. Salakhutdinov, Science **313**, 504 (2006), doi:10.1126/science.1127647

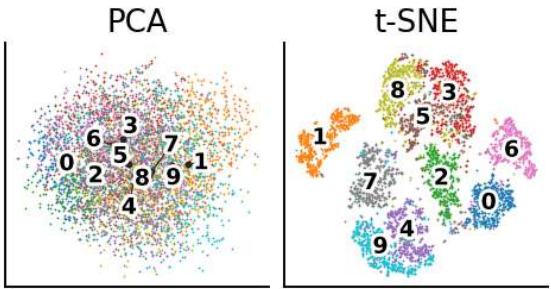
The Target Is the Input Itself



$$\varepsilon(\mathbf{x}) = \frac{1}{D}\sum_{j=1}^{D}\left(x_j - \tilde{x}_j\right)^2, \qquad \mathcal{L}(\boldsymbol{\theta}_E, \boldsymbol{\theta}_D) = \frac{1}{n}\sum_{i=1}^{n}\varepsilon(\mathbf{x}_i)$$

right panel: $(x_j - \tilde{x}_j)^2$ per pixel; ε: reconstruction error; a linear autoencoder finds the PCA subspace (Baldi, Hornik) P. Baldi, K. Hornik, Neural Networks **2**, 53 (1989), doi:10.1016/0893-6080(89)90014-2

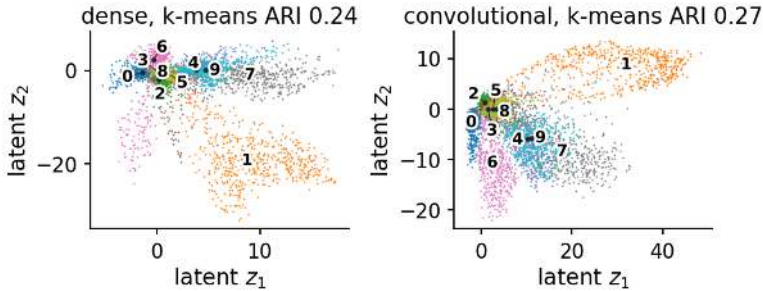
Similar Digits Get Nearby Latent Vectors



16 latent numbers of 5000 test digits, in 2D; colours: true digits (plotting only)

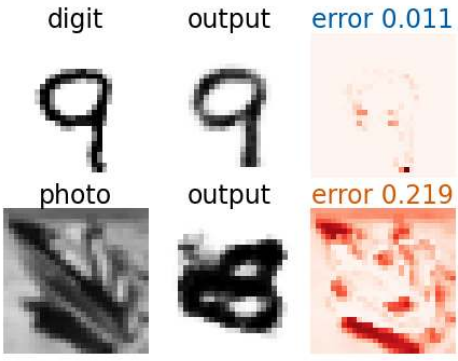
- 784 pixels $\rightarrow$ $d = 16$ numbers
- PCA of $\mathbf{z}$: the digits overlap
- t-SNE of $\mathbf{z}$: the groups appear

Dense and Convolutional Encoders with Two Latent Numbers



- $d = 2$: the latent space is drawn directly; colours: true digits
- test error per pixel: dense 0.039; convolutional (compares neighbouring pixels) 0.045
- k-means ($k = 10$) on $\mathbf{z}$ against the digits, adjusted Rand index (1 identical, 0 chance): 0.24 and 0.27

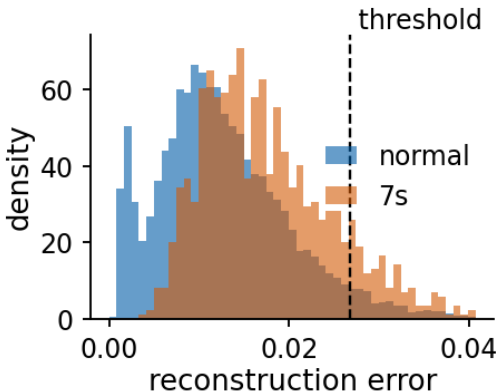
Anomaly Detection with an Autoencoder



trained on all digits: a typical digit and a typical photograph (median errors)

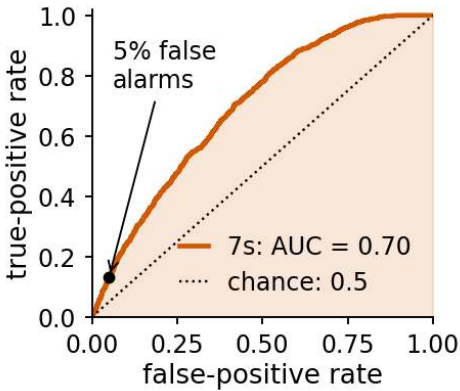
- data like the training data: small $\varepsilon(\mathbf{x})$
- unlike them: **large** $\varepsilon(\mathbf{x})$, an **anomaly**
- rule: flag ε above a threshold, e.g. 5% false alarms

The Digit 7 Is a Subtle Anomaly



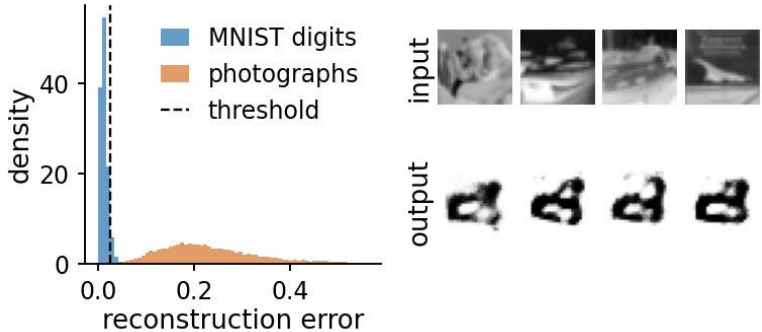
- trained on all digits *except* 7; histogram of ε: trained digits (blue), 7s (orange)
- threshold (dashed, 5% false alarms) flags 13% of the 7s
- a 7 shares strokes with 1 and 9

ROC Curve and AUC



- each threshold: true-positive rate (TPR, anomalies found) against false-positive rate (FPR, false alarms)
- receiver operating characteristic (ROC): all thresholds; diagonal = chance
- AUC, area under the curve: 0.5 chance, 1 perfect; 0.70 for the 7s

Photographs Are a Clear Anomaly



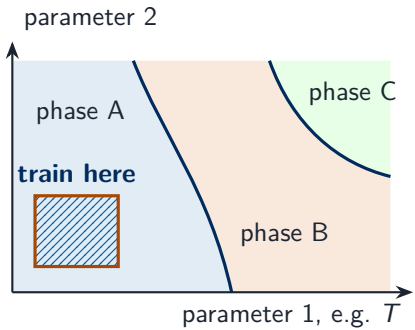
- CIFAR-10 photographs (28×28, grayscale) into the digit autoencoder
- median error 0.219 (digits 0.011); 99.8% flagged; AUC 0.9997
- rebuilt as blurred digits: the decoder produces what it learned

5

Anomaly Detection for Studies of Phase Diagrams

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Anomaly Detection on a Phase Diagram



- **train** on one well-understood region (hatched)
- **scan** the reconstruction error over the whole diagram
- **boundaries**: where the error changes fastest

schematic; thick lines: phase boundaries

Used for the one-dimensional extended Bose–Hubbard model by Kottmann et al.

K. Kottmann, P. Huembeli, M. Lewenstein, A. Acín, Phys. Rev. Lett. **125**, 170603 (2020),
doi:10.1103/PhysRevLett.125.170603

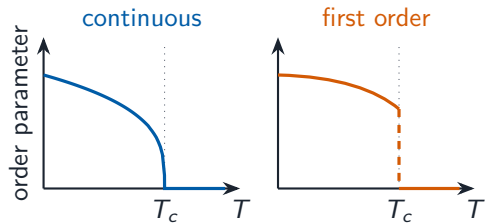
Three Models, Physics First

	model	what makes it interesting
1	2D Ising	one transition, exactly known: the reference
2	J_1–J_2 Ising	competing couplings; m misses a whole phase
3	2D XY	no local order parameter; a vortex transition

For each: Hamiltonian, order parameter, phases $\rightarrow$ phase diagram $\rightarrow$ PCA and autoencoder.

Continuous and First-Order Phase Transitions

- **continuous**: order parameter falls to zero; ξ diverges
- **first order**: order parameter jumps; phases coexist
- XY model: infinite-order Berezinskii–Kosterlitz–Thouless (BKT) transition

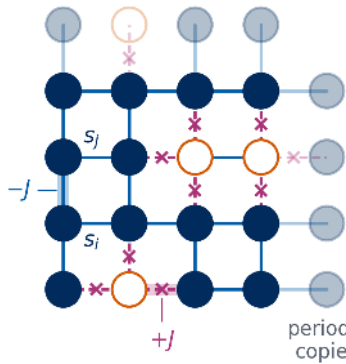


Schematic order parameter against temperature; ξ: correlation length.

Model 1

The 2D Ising Model

The 2D Ising Model and Its Hamiltonian



$$H(\mathbf{s}) = -J \sum_{\langle i,j \rangle} s_i s_j$$

$J > 0$, $\langle i, j \rangle$: nearest neighbours
periodic boundaries (a torus)

$s_i = +1$

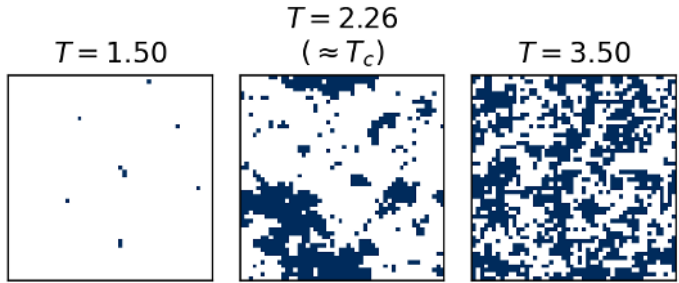
$s_i = -1$

satisfied bond: $-J$

broken bond: $+J$

$p(\mathbf{s}) = e^{-\beta H(\mathbf{s})}/Z$, $\beta = 1/T$ ($k_B = J = 1$), $N = L^2$ spins; $H(-\mathbf{s}) = H(\mathbf{s})$: flipping every spin costs nothing

Ordered and Disordered Phases

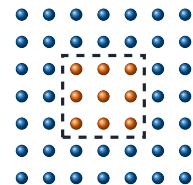


- order parameter: magnetisation per spin $m = \frac{1}{N}\sum_i s_i$
- $T < T_c$: $m = \pm M(T)$, spontaneous magnetisation $M > 0$; the symmetry $\mathbf{s} \to -\mathbf{s}$ is broken
- $T > T_c$: small domains, $m \to 0$ for $L \to \infty$

Domain-Wall Energy against Entropy

equilibrium minimises $F = E - TS$

- 1D: one wall changes F by $2J - T\ln(N-1)$, negative for long chains: no order at $T > 0$
- 2D: a wall of length ℓ changes F by about $\ell\,(2J - T\ln 3)$
- order survives below $2J/\ln 3 = 1.82\,J$, a lower bound (Peierls); exact $T_c = 2.269\,J$



wall of length $\ell = 12$

R. Peierls, Proc. Camb. Phil. Soc. **32**, 477 (1936), doi:10.1017/S0305004100019174

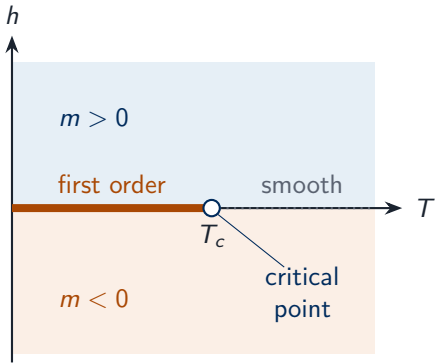
A Continuous Transition at $T_c = 2.269$

$$T_c = \frac{2}{\ln(1+\sqrt{2})} = 2.269 \text{ (Onsager)}, \qquad t = \frac{T - T_c}{T_c}$$

- $m \sim (-t)^{1/8}$ below T_c
- correlation length $\xi \sim |t|^{-1}$, susceptibility $\chi \sim |t|^{-7/4}$, specific heat $c \sim -\ln|t|$
- exponents shared by the **2D Ising universality class**

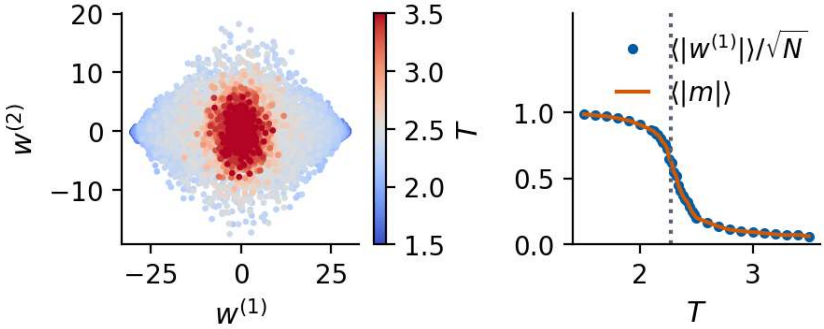
L. Onsager, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117; C. N. Yang, Phys. Rev. **85**, 808 (1952), doi:10.1103/PhysRev.85.808

Phase Diagram in the (T, h) Plane



- field h:
  $H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$
- $h = 0$, $T < T_c$: **first-order** line, m jumps; it ends at the **critical point**
- our data: $h = 0$, a scan in T

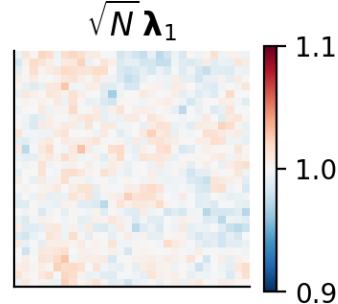
PCA Finds the Order Parameter



- 9472 configurations, 30×30, 37 temperatures; PCA sees only the spins
- left: $(w_i^{(1)}, w_i^{(2)})$, $w_i^{(l)} = \boldsymbol{\lambda}_l^\top(\mathbf{s}_i - \bar{\mathbf{s}})$, coloured by T
- right: $\langle|w^{(1)}|\rangle_T/\sqrt{N}$, the mean over the configurations at T, on top of $\langle|m|\rangle$

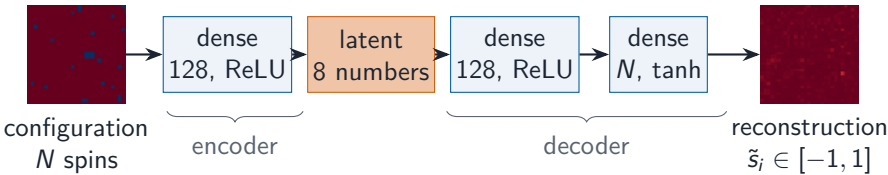
The First Principal Component Is the Magnetisation

- below T_c: $\langle s_i s_j \rangle \approx M^2$, so $\boldsymbol{\lambda}_1 \approx (1, \ldots, 1)/\sqrt{N}$
- $w^{(1)} \approx \frac{1}{\sqrt{N}} \sum_i s_i = \sqrt{N}\, m$
- measured: spread of $\sqrt{N}\,\boldsymbol{\lambda}_1$ 1%; correlation of $|w^{(1)}|$ with $|m|$ 0.99996



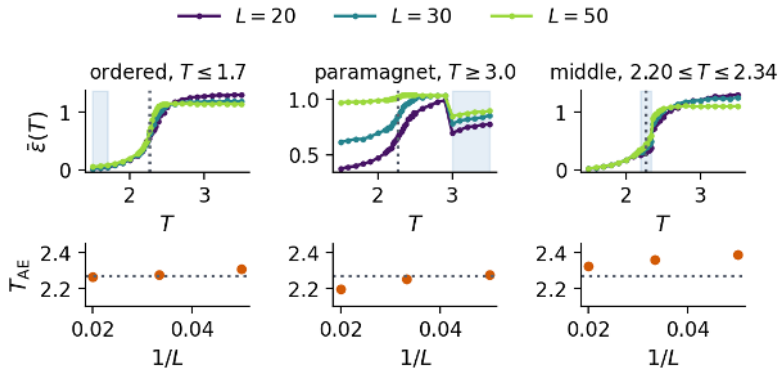
m is linear in the spins; a linear method finds it.

Autoencoder on Spin Configurations



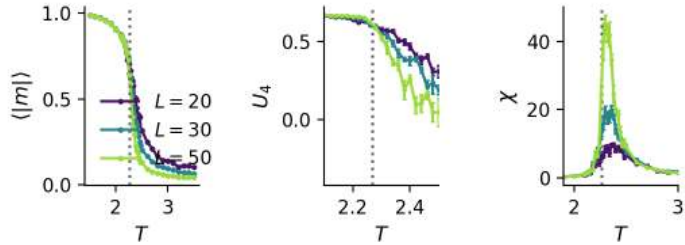
- $L = 20, 30, 50$; trained only on the 768 configurations with $T \leq 1.7$
- error $\varepsilon(\mathbf{s}) = \frac{1}{N}\sum_i (s_i - \tilde{s}_i)^2$; mean $\bar{\varepsilon}(T)$ at each temperature
- step fit $\bar{\varepsilon}(T) = a + b \tanh\Big((T - T_{\mathrm{AE}})/\delta\Big)$: centre T_{AE} (AE: autoencoder), width δ

The Training Region Decides the Signal



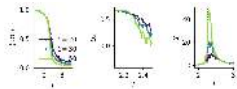
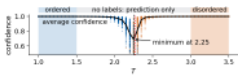
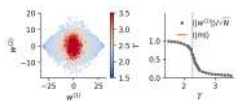
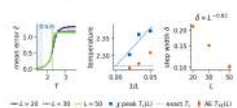
- top: $\bar{\varepsilon}(T)$, training window shaded; bottom: T_{AE} (labelled T^*) against $1/L$
- ordered: $T_{\mathrm{AE}} = 2.308, 2.278, 2.264 \to T_c$; paramagnet: nearly flat; middle: step at the window edge

Physics Needed beyond the Ising Snapshots



- on one lattice PCA and the autoencoder show a smooth crossover
- Binder cumulant $U_4 = 1 - \langle m^4 \rangle / 3 \langle m^2 \rangle^2$: curves of different L cross at 2.2495 and 2.2784; peaks of the susceptibility $\chi = N\beta(\langle m^2 \rangle - \langle |m| \rangle^2)$ extrapolate to $T_c = 2.2694$
- $w^{(1)} = \sqrt{N} m$ is recognised only by comparison with the known m

Four Ways to Find T_c

	method	what we gave it	what it found
	finite-size scaling	the order parameter m	$T_c = 2.2694$
	CNN with labels	labels far from T_c	lowest confidence at 2.25
	PCA	spins only	the magnetisation
	autoencoder	examples of one phase	$T_{\mathrm{AE}} = 2.2641$ at $L = 50$

The less we tell the method, the less of the answer comes from us.

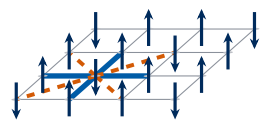
Model 2

A Frustrated Magnet: the J_1–J_2 Ising Model

The J_1–J_2 Model: Two Couplings that Compete

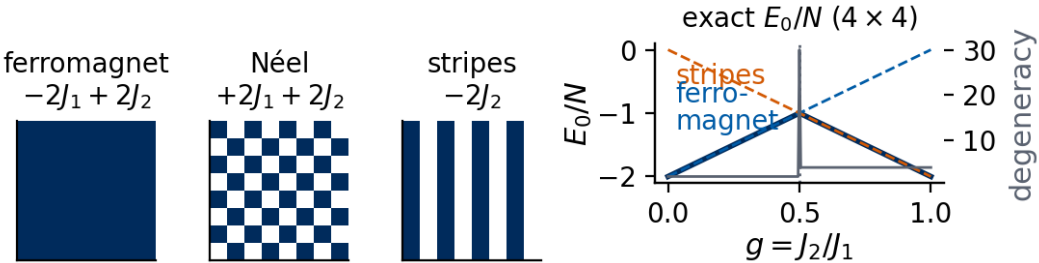
$$H = -J_1 \sum_{\langle i,j \rangle} s_i s_j \; + \; J_2 \sum_{\langle\langle i,j \rangle\rangle} s_i s_j, \qquad s_i = \pm 1, \qquad g = \frac{J_2}{J_1}$$

- $J_1 > 0$, 4 nearest neighbours: **ferromagnetic**
- $J_2 > 0$, 4 diagonal neighbours: **antiferromagnetic**
- no square of four spins satisfies all six bonds: **frustration**



J_1, nearest (solid)
J_2, diagonal (dashed)

Ground States: Ferromagnet or Stripes

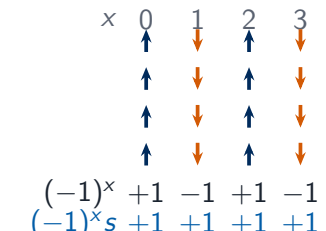


- $E/N = -2J_1 + 2J_2$ (ferromagnet), $-2J_2$ (stripes), $2J_1 + 2J_2$ (Néel)
- ferromagnet for $g < 1/2$, stripes for $g > 1/2$; at $g = 1/2$ the 4×4 torus has 30 ground states instead of $2 + 4$
- exact limits: Ising at $g = 0$, $T_c = 2.269\, J_1$; $T_c \to 2.269\, J_2$ for $g \to \infty$

Fourier Components of a Spin Configuration

$$m_{\mathbf{k}} = \frac{1}{N} \sum_{\mathbf{r}} s_{\mathbf{r}}\, e^{i\mathbf{k}\cdot\mathbf{r}}, \qquad \mathbf{r} = (x, y)$$

- $\mathbf{k} = (0, 0)$: $m_{(0,0)} = m$, the magnetisation
- $\mathbf{k} = (\pi, 0)$: factor $(-1)^x$; vertical stripes give $m_{(\pi,0)} = \pm 1$, $m = 0$
- $m_{\mathrm{str}} = \left(m_{(\pi,0)}^2 + m_{(0,\pi)}^2\right)^{1/2} = 1$ for either stripe orientation



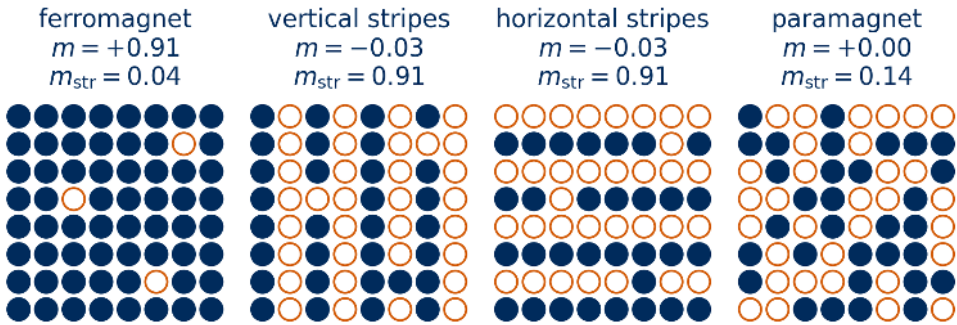
vertical stripes, $s_{\mathbf{r}} = (-1)^x$

Fourier Components Are the Order Parameters

phase	$\mathbf{k}$	order parameter	where
ferromagnet	$(0,0)$	$m = \frac{1}{N}\sum_{\mathbf{r}} s_{\mathbf{r}}$	$g < 1/2$, low T
stripes	$(\pi,0),(0,\pi)$	$m_{\mathrm{str}} = \sqrt{m^2_{(\pi,0)} + m^2_{(0,\pi)}}$	$g > 1/2$, low T
paramagnet	–	$m \approx 0,\ m_{\mathrm{str}} \approx 0$	high T

Stripe phase: $m = 0$ **although the system is ordered**.

Three Phases and Costless Domain Walls at $g = 1/2$



- the cheapest domain wall costs $2J_1|1 - 2g|$ per unit length at $T = 0$
- at $g = 1/2$ walls cost nothing: the ordering temperature **drops to zero**

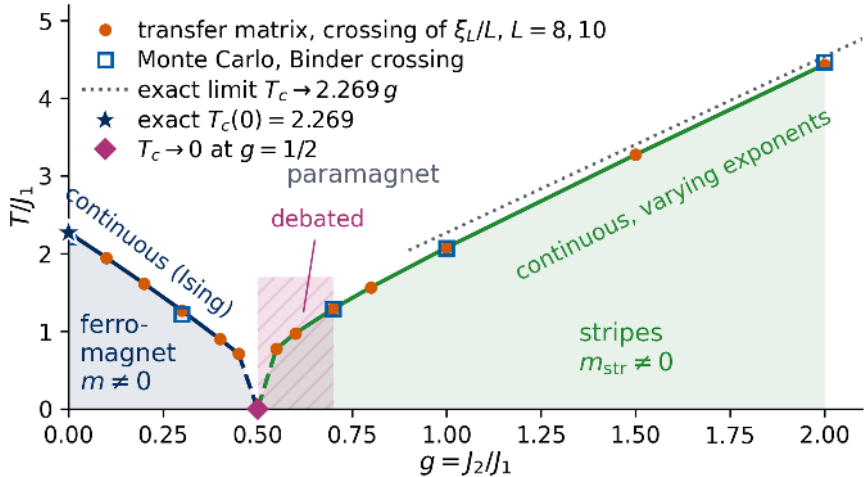
The Transition Changes Character along the Boundary

- $g < 1/2$: continuous, 2D Ising class; $T_c(g) \to 0$ as $g \to 1/2$
- $g > 1/2$: close to $1/2$ debated, weakly first order up to g^* in Monte Carlo; then continuous, varying exponents
- Monte Carlo: $g^* = 0.67(1)$ (Jin et al.); tensor networks: $g^* \simeq 0.54$ (Li and Yang)



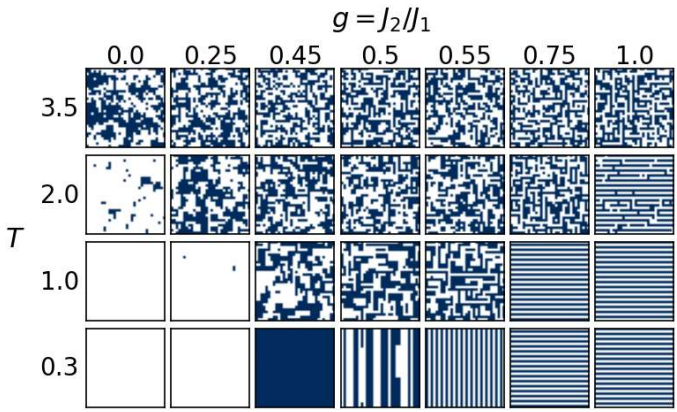
D. P. Landau, K. Binder, Phys. Rev. B **31**, 5946 (1985), doi:10.1103/PhysRevB.31.5946; A. Kalz, A. Honecker, S. Fuchs, T. Pruschke, Eur. Phys. J. B **65**, 533 (2008), arXiv:0805.0983; A. Kalz, A. Honecker, M. Moliner, Phys. Rev. B **84**, 174407 (2011), doi:10.1103/PhysRevB.84.174407; S. Jin, A. Sen, A. W. Sandvik, Phys. Rev. Lett. **108**, 045702 (2012), arXiv:1110.5874; S. Jin, A. Sen, W. Guo, A. W. Sandvik, Phys. Rev. B **87**, 144406 (2013), arXiv:1212.5339; H. Li, L.-P. Yang, Phys. Rev. E **104**, 024118 (2021), arXiv:2103.09464

Phase Diagram in the (g, T) Plane



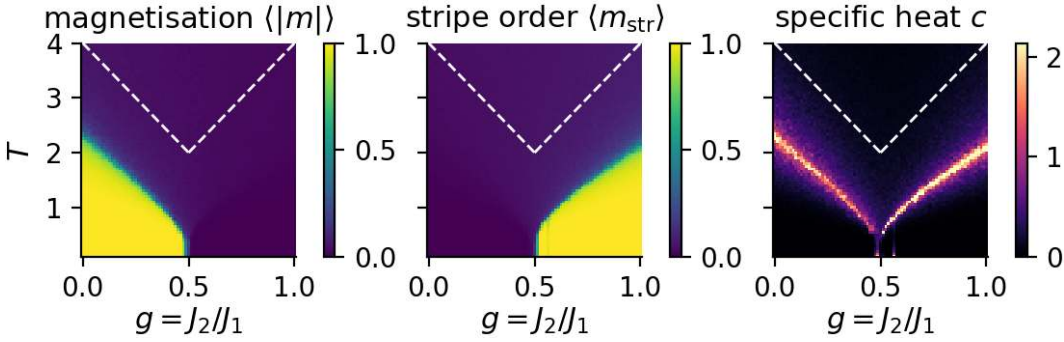
ξ_L/L: correlation length of an infinite strip of width L over L, from transfer matrices; hatched: transition type debated

Snapshots across the (g, T) Plane



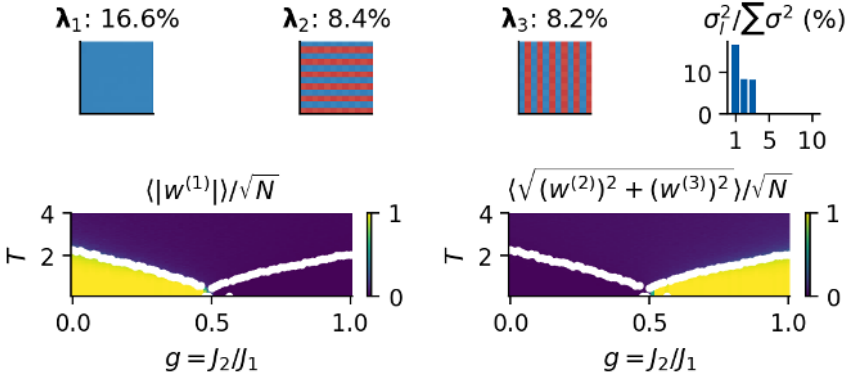
- 32×32, Metropolis
- 81 values of g, 79 temperatures
- 100 snapshots per point

The Observable Decides What We See



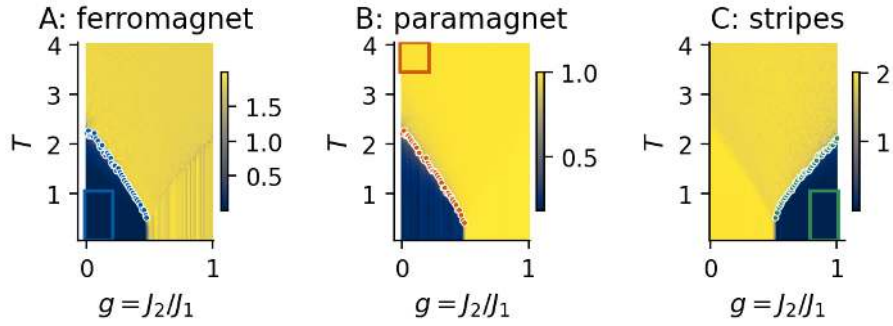
- $\langle|m|\rangle$: only the ferromagnet; the stripe phase looks like a paramagnet
- $\langle m_{\mathrm{str}}\rangle$: the stripes, because we knew which pattern to measure
- $c = N\beta^2(\langle e^2\rangle - \langle e\rangle^2)$, $e = H/N$: peaks at every transition; dashed: mean field

PCA Finds Both Order Parameters



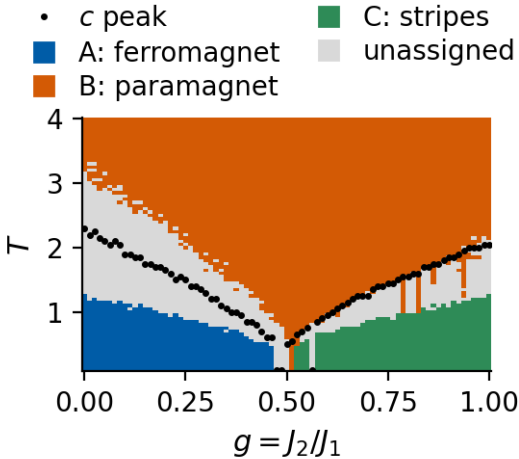
- 127 980 snapshots; top: $\boldsymbol{\lambda}_{1,2,3}$, with 16.6%, 8.4%, 8.2% of the variance
- bottom: $\langle|w^{(1)}|\rangle/\sqrt{N} \approx \langle|m|\rangle$, $\langle\sqrt{(w^{(2)})^2+(w^{(3)})^2}\rangle/\sqrt{N} \approx \langle m_{\mathrm{str}}\rangle$

Three Training Regions Rebuild Different Phases



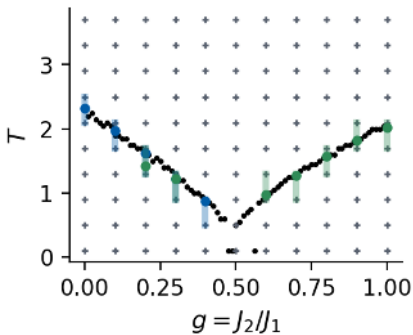
- convolutional autoencoder, 4 latent numbers, trained on one corner (rectangle), with flipped copies $-\mathbf{s}$
- A: stripes and paramagnet both "not ferromagnet"; B: stripes **invisible**; C: the stripe boundary

Combining the Regions into a Phase Map



- score $a_\rho = |\bar{\varepsilon}_\rho - \bar{\varepsilon}_\rho^{\mathrm{tr}}|/\mathrm{sd}_\rho$, $\rho = \mathrm{A}, \mathrm{B}, \mathrm{C}$; $\bar{\varepsilon}_\rho^{\mathrm{tr}}$, sd_ρ: mean and standard deviation of $\bar{\varepsilon}_\rho$ on region ρ
- lowest score wins; *unassigned* if all exceed 3 (21% of the grid)
- boundaries a median 0.025 from the specific-heat peak

A Coarse Scan Tells Where to Measure



- coarse grid, $\Delta g = 0.1$, $\Delta T = 0.4$, 10 snapshots per point
- bars: brackets of the error jump; dots: dense scan inside them
- 3260 snapshots, **0.5%** of the dense map

Physics Needed beyond the J_1–J_2 Snapshots

- PCA and the autoencoder locate changes; they do not give the transition type
- energy histogram at the transition: one peak continuous, two peaks first order (Kalz et al., $g = 0.6$)
- near g^* the type needs much larger lattices

A. Kalz, A. Honecker, S. Fuchs, T. Pruschke, Eur. Phys. J. B **65**, 533 (2008), arXiv:0805.0983

What Each Method Needs and Finds

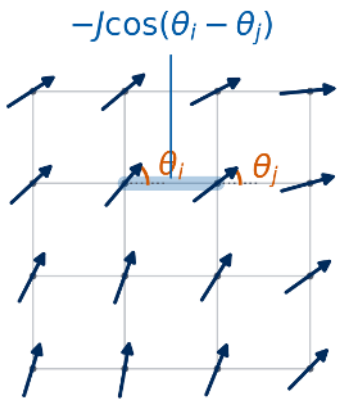
method	input	result
$\langle \lvert m \rvert \rangle$ scan	ferromagnetic order parameter	misses the stripe phase
specific heat	many samples per point	both boundaries; expensive
PCA	all snapshots	both order parameters
AE, one region	snapshots of one corner	the boundary of that phase
AE, three regions	three corners	three phases; coarse scan at 0.5%

The method finds where something changes; which order sets in is the physicist's question.

Model 3

The 2D XY Model

Planar Spins and the XY Hamiltonian



$$H = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j)$$

planar spins $\mathbf{s}_i = (\cos\theta_i, \sin\theta_i)$

$\theta_i \in [0, 2\pi)$, $J > 0$

$\langle i,j \rangle$: nearest neighbours



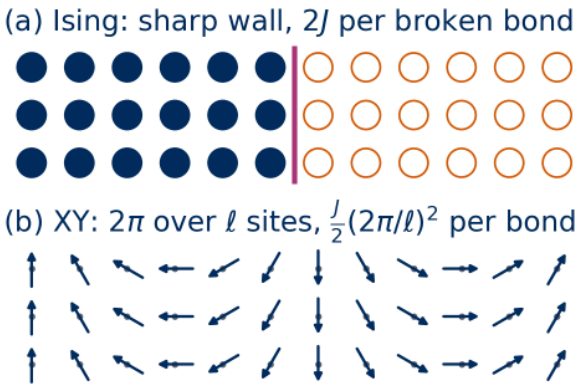
parallel: $-J$

antiparallel: $+J$

small angle δ: $-J + \frac{J}{2}\delta^2$

$N = L^2$ angles θ_i on an $L \times L$ lattice, periodic boundaries; $\langle i,j \rangle$: the $2N$ nearest-neighbour bonds; $J = k_B = 1$

Slow Twists Destroy the Magnetisation



- a 2π twist over ℓ sites costs about the same for every ℓ in 2D
- $\langle|\mathbf{m}|\rangle \to 0$ as $L \to \infty$ at every $T > 0$ (Mermin–Wagner), $\mathbf{m} = \frac{1}{N}\sum_i \mathbf{s}_i$
- slow decay: $\langle|\mathbf{m}|\rangle = 0.940,\ 0.907,\ 0.874$ for $L = 8, 32, 128$ $(T = 0.3)$

N. D. Mermin, H. Wagner, Phys. Rev. Lett. **17**, 1133 (1966), doi:10.1103/PhysRevLett.17.1133

The Berezinskii–Kosterlitz–Thouless Transition

vortex: the spin direction turns by $\pm 360^\circ$ around a point

below T_{BKT}

bound vortex–antivortex pairs
$G(r) = \langle \cos(\theta_0 - \theta_r) \rangle$: power law
stiffness $\Upsilon > 0$

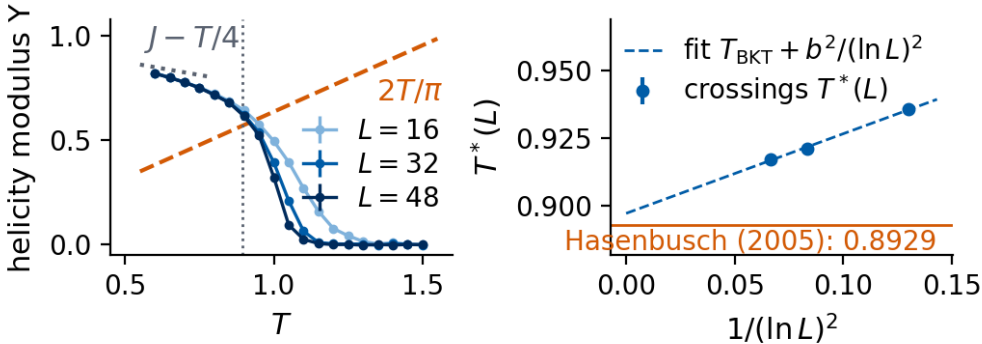
above T_{BKT}

free vortices (entropy wins)
$G(r)$: exponential
$\Upsilon = 0$

No order parameter; Υ jumps from $2T/\pi$ to 0 at $T_{\mathrm{BKT}} = 0.893\,J$.

V. L. Berezinskii, Sov. Phys. JETP **32**, 493 (1971), **34**, 610 (1972); J. M. Kosterlitz, D. J. Thouless, J. Phys. C **6**, 1181 (1973), doi:10.1088/0022-3719/6/7/010; M. Hasenbusch, J. Phys. A **38**, 5869 (2005), doi:10.1088/0305-4470/38/26/003; 2016 Nobel Prize in Physics to Kosterlitz and Thouless (with Haldane), nobelprize.org/prizes/physics/2016/summary

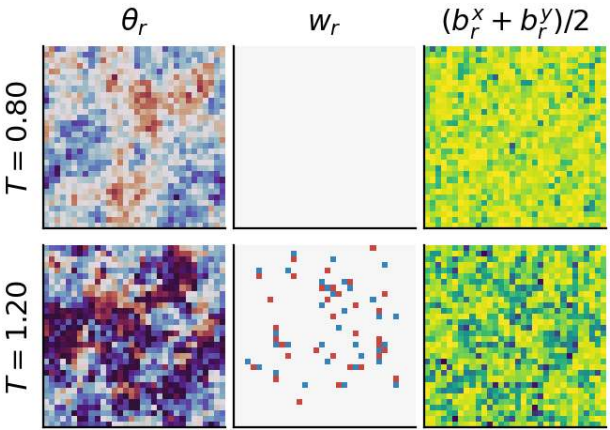
The Stiffness Drops to Zero at T_{BKT}



- left: stiffness (helicity modulus) Υ, the cost of a twist Δ, $F(\Delta) - F(0) = \frac{\Upsilon}{2}\Delta^2$
- right: crossings $T^*(L)$ of Υ with $2T/\pi$ (b: fit constant) $\rightarrow 0.897 \pm 0.003$

D. R. Nelson, J. M. Kosterlitz, Phys. Rev. Lett. **39**, 1201 (1977), doi:10.1103/PhysRevLett.39.1201

Three Representations of a Snapshot



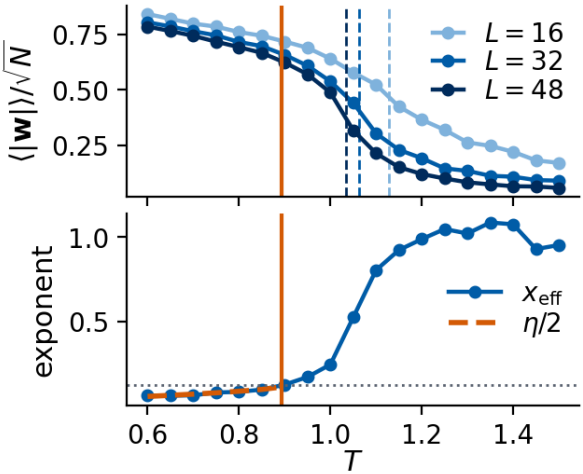
- **spins** $(\cos\theta_r, \sin\theta_r)$
- **vortices** w_r: winding number per plaquette
- **bonds** $b_r^\mu = \cos(\theta_{r+\hat{\mu}} - \theta_r)$: local energy

$L = 16, 32, 48$, 19 temperatures 0.60–1.50, 192 snapshots each

one $L = 32$ snapshot per row

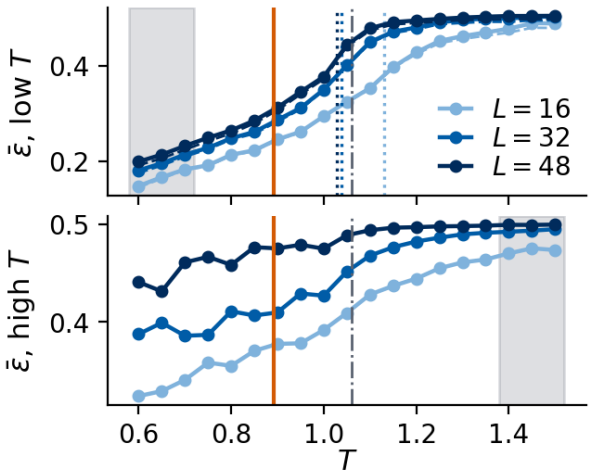
M. J. S. Beach, A. Golubeva, R. G. Melko, Phys. Rev. B **97**, 045207 (2018), doi:10.1103/PhysRevB.97.045207

PCA Follows the Finite-Lattice Magnetisation



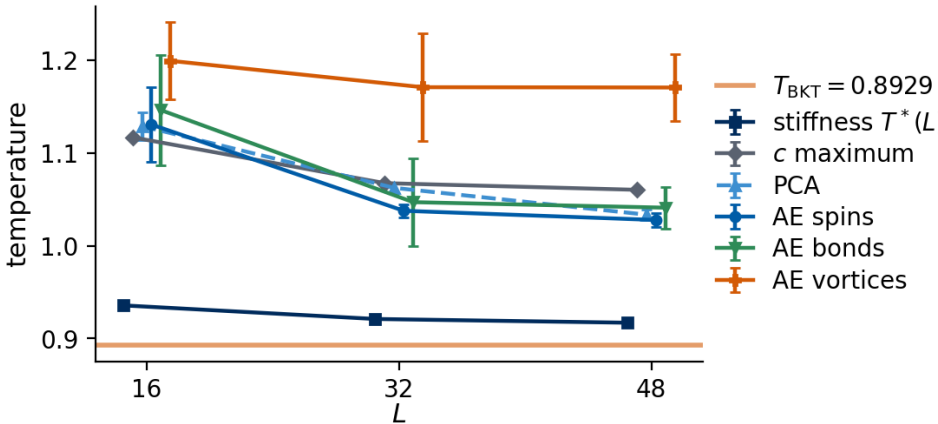
- two uniform leading directions: $\langle|\mathbf{w}|\rangle/\sqrt{N} \approx \langle|\mathbf{m}|\rangle$
- top: steepest fall (dashed) at 1.13, 1.06, 1.03 for $L = 16, 32, 48$; orange: T_{BKT}
- bottom: x_{eff} in $\langle|\mathbf{w}|\rangle \propto L^{-x_{\mathrm{eff}}}$, close to $\eta/2$ below T_{BKT}, with $G(r) \propto r^{-\eta}$

The Autoencoder Learns the Magnetisation



- 4 latent numbers; training window grey; top: trained at low T, bottom: at high T
- top: $\bar{\varepsilon}$ follows $(1 - |\mathbf{m}|^2)/2$ (dashed); steepest rise (dotted) at 1.13, 1.04, 1.03
- orange: T_{BKT}; dash-dotted: specific-heat maximum, $L = 48$

Unsupervised Estimates against the Known Temperatures



$L = 48$: estimates 1.03–1.17, **0.14–0.28 above** T_{BKT}; bonds give the specific-heat maximum, vortex maps 1.17; c: specific heat per spin

Physics Needed beyond the XY Snapshots

- snapshots show the finite-lattice magnetisation crossover, above T_{BKT}
- the stiffness locates T_{BKT}: crossings $T^{*}(L) = 0.936,\ 0.921,\ 0.917$ $\rightarrow 0.897 \pm 0.003$ (literature 0.8929)
- crossovers extrapolated in $1/(\ln L)^2$: 0.932 ± 0.020 (PCA), 0.952 ± 0.036 (AE)

Three Models, One Protocol

model	order	PCA and autoencoder on snapshots	what locates the transition
2D Ising	m, linear in the spins	PC 1 $\propto m$; $T_{\mathrm{AE}} = 2.2641$ at $L = 50$	finite-size scaling
J_1–J_2 Ising	m and m_{str}, linear in the spins	both order parameters; boundaries of the training phases	several training regions
2D XY	no local order parameter	magnetisation crossover, ≈ 1.03 at $L = 48$	stiffness Υ: T_{BKT}

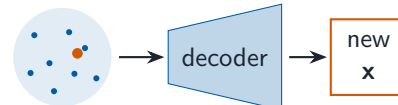
Summary

- **unsupervised learning**: few coordinates (PCA, t-SNE), groups, and unusual samples (autoencoders), without labels
- **PCA** finds order parameters that are linear in the spins: m (Ising), m and m_{str} (J_1–J_2)
- **anomaly detection** marks where snapshots stop resembling the training phase; which order sets in, and the transition type, need physics

Learning a Probability Distribution from Data

The data are samples from an unknown $p(\mathbf{x})$; we learn p itself and draw *new* samples from it.

- maximum likelihood, and latent-variable models
- variational autoencoders: new digits from random latent vectors
- conditional generation: a chosen digit, or a spin configuration at a chosen temperature



A latent-variable model samples in two steps, first $\mathbf{z}$ from a simple distribution, then $\mathbf{x}$ from the decoder.

For the Ising model we know the true distribution, $p(\mathbf{s}) = e^{-H(\mathbf{s})/T}/Z$: a test of what a network has learned.

Thank You!

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar



6
Clustering



JAGIELLONIAN UNIVERSITY
IN KRAKOW

k-means Clustering

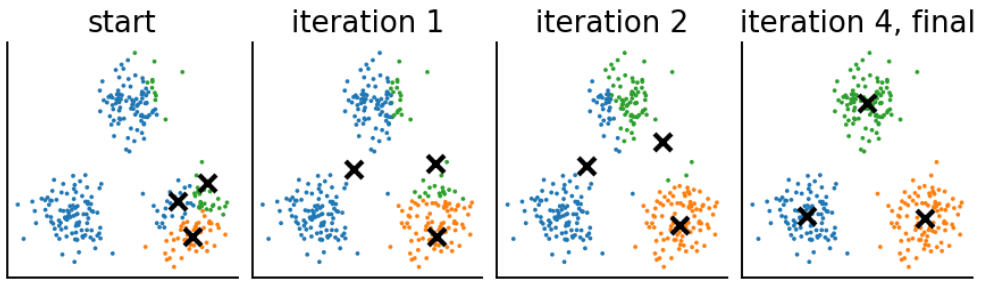
$$\mathcal{L} = \sum_{c=1}^{k} \sum_{i \in S_c} \|\mathbf{x}_i - \boldsymbol{\mu}_c\|^2$$

S_c: cluster c; $\boldsymbol{\mu}_c$: its centre

1. place k centres at random
2. **assign** each sample to its nearest centre; **update** each centre to the mean of its samples
3. repeat until nothing changes: $\mathcal{L}$ never increases, so it stops, possibly in a poor minimum

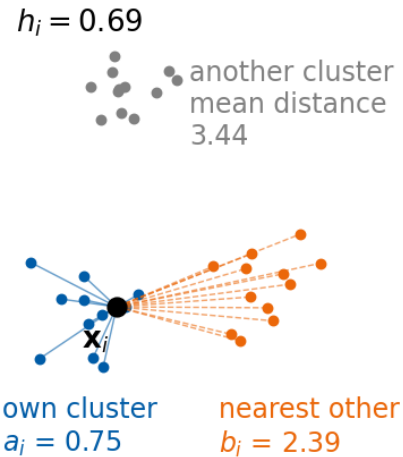
S. P. Lloyd, IEEE Trans. Inf. Theory **28**, 129 (1982), doi:10.1109/TIT.1982.1056489

k-means, Iteration by Iteration



Three groups of points in 2D, the three centres (crosses) placed badly on purpose; colours show the assignment. The loss never increases, and after a few iterations nothing changes.

The Silhouette Score



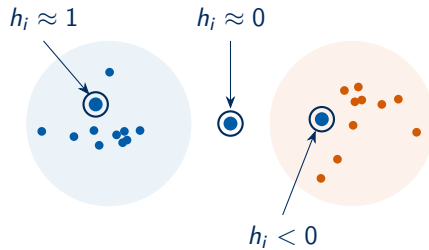
- a_i: mean distance to the other samples of its cluster
- b_i: mean distance to the nearest other cluster

$$h_i = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1, 1]$$

P. J. Rousseeuw, J. Comput. Appl. Math. **20**, 53 (1987), doi:10.1016/0377-0427(87)90125-7

The Mean Silhouette Chooses the Number of Clusters

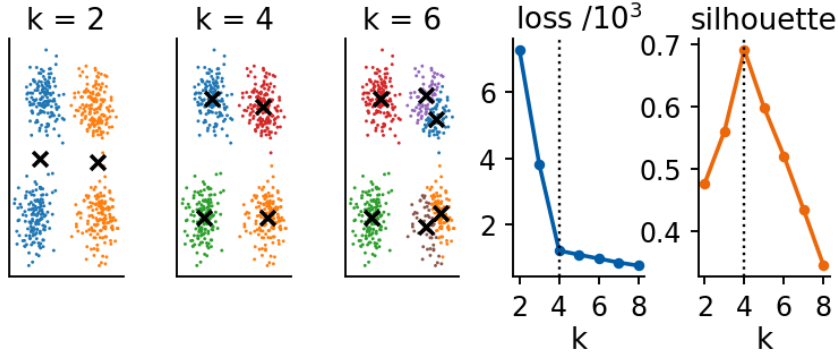
- $h_i \approx 1$: well inside its cluster, far from the others
- $h_i \approx 0$: on the border between two clusters
- $h_i < 0$: closer to another cluster, probably assigned wrongly



Schematic. The three circled samples belong to the blue cluster.

$\bar{h}(k) = \frac{1}{n} \sum_i h_i$ for k clusters; choose the k with the largest $\bar{h}(k)$.

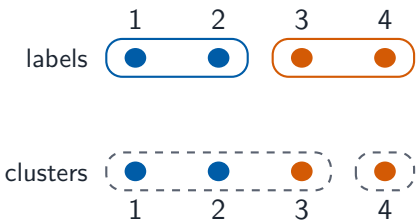
Elbow and Silhouette Both Point to Four Clusters



- four groups of points; k-means for $k = 2, \ldots, 8$, 10 random starts each
- elbow: the loss $\mathcal{L}(k)$ bends at $k = 4$
- silhouette: $\bar{h}(k)$ highest at $k = 4$, 0.69 against 0.56 ($k = 3$) and 0.60 ($k = 5$)

Rand Index

To score how well a clustering recovers known classes, we compare two partitions of the same n samples, the true labels and the clusters, *pair by pair*; the names of the groups do not matter.

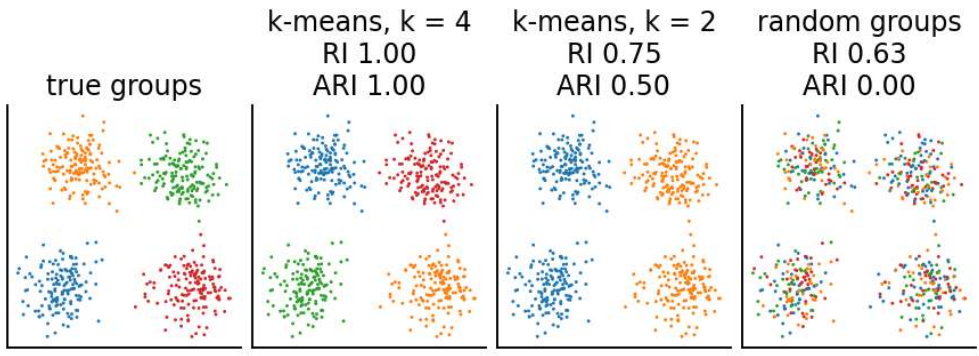


pair	labels	clusters	agree
1–2	same	same	✓
1–3	different	same	×
1–4	different	different	✓
2–3	different	same	×
2–4	different	different	✓
3–4	same	different	×

A pair **agrees** if both partitions put it in the same group, or both in different groups. The **Rand index** is the fraction of agreeing pairs, out of all $n(n-1)/2$:

$$\mathrm{RI} = \frac{\text{agreeing pairs}}{n(n-1)/2} = \frac{3}{6} \quad \text{here;} \qquad \mathrm{RI} = 1 \text{ for identical partitions}$$

Random Groups Already Reach a High Rand Index



Random groups already reach a high RI (0.63 here): most pairs are in different groups in both partitions.

Adjusted Rand Index

- the adjusted Rand index corrects for chance:

$$\mathrm{ARI} = \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{1 - \mathbb{E}[\mathrm{RI}]},$$

  with $\mathbb{E}[\mathrm{RI}]$ the mean RI when the samples are reassigned at random, keeping the group sizes

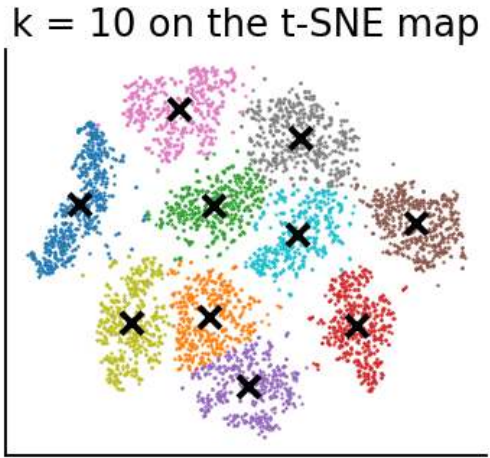
- **ARI = 1**: the same partition; **ARI $\approx$ 0**: no better than chance; it can be negative

partition	RI	ARI
k-means, $k = 4$	0.998	0.996
k-means, $k = 2$	0.748	0.495
random groups	0.626	0.001

600 points around four centres and three partitions of them

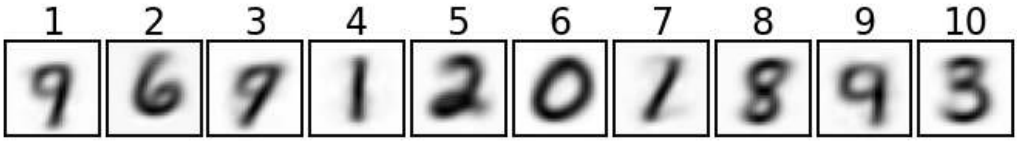
L. Hubert and P. Arabie, *Comparing partitions*, J. Classification **2**, 193 (1985), doi:10.1007/BF01908075

k-means on MNIST



- k-means with $k = 10$ on the t-SNE map of 5000 digits
- crosses: the ten centres

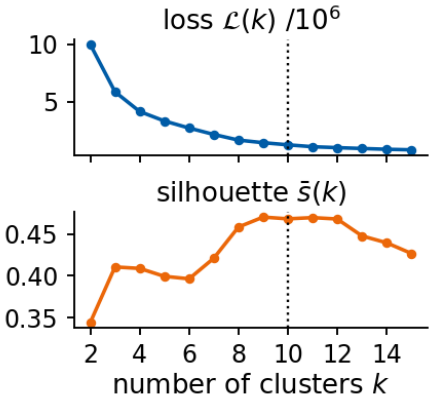
Ten Centres and the Agreement with the Digits



the ten centres $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_{10}$ found on 50 PCA components, shown as images

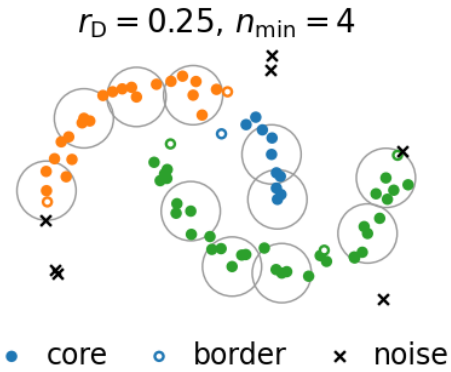
Adjusted Rand index with the true digits: 0.38 on 50 PCA components, 0.76 on the t-SNE map.

Choosing k on MNIST



- loss $\mathcal{L}(k)$ and mean silhouette $\bar{h}(k)$ (labelled $\bar{s}$) on the t-SNE map; dotted: $k = 10$
- no sharp elbow; $\bar{h}$ highest at $k = 9$, almost equal at 10 and 11
- on real data the guides suggest a range of k

DBSCAN: Clustering by Density



density-based spatial clustering of applications with noise (DBSCAN); radius r_D, minimum number of points n_{min}

- **core point:** at least n_{min} points within r_D (grey circles)
- **border point:** within r_D of a core point
- **noise:** all other points (crosses)

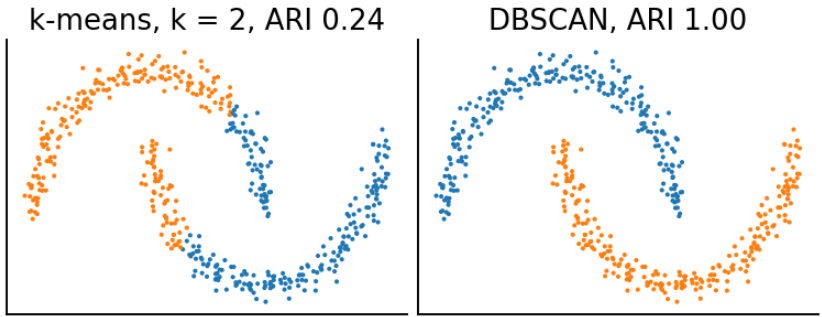
M. Ester, H.-P. Kriegel, J. Sander, X. Xu, KDD 1996, AAAI proceedings

The DBSCAN Algorithm

1. count the neighbours within r_{D} of every point (itself included); mark the core points
2. start a new cluster at a core point not yet in a cluster
3. add every point within r_{D} of a core point of the cluster, until nothing is added
4. repeat 2 and 3; points never reached are noise

$r_{\mathrm{D}} = 0.25$, $n_{\mathrm{min}} = 4$: 3 clusters, 65 core, 6 border and 7 noise points; no number of clusters is needed.

DBSCAN Finds Clusters of Any Shape



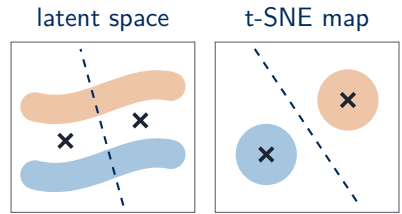
- two half-moons, 500 points
- k-means cuts the space into convex cells, so it finds round clusters: with $k = 2$ it gives ARI 0.24
- DBSCAN follows the dense regions: ARI 1.00

The Latent Space Keeps the Digits in Elongated Groups

k-means ($k = 10$) against the true digits, adjusted Rand index (ARI):

	directly	t-SNE map
50 PCA components	0.38	0.76
16 latent numbers	0.28	0.70

- the digits are kept in the latent space, but not as the round groups that k-means looks for
- t-SNE only *draws* the latent space; the latent space itself takes *new* images (encoder) and gives back images (decoder)



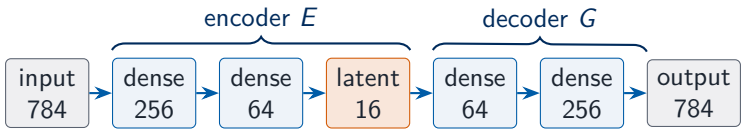
Schematic with two groups. k-means splits the space by a straight boundary (dashed) between its centres (crosses). It cuts elongated groups across and separates compact ones.

7

Autoencoders: More Details

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Dense Autoencoder: Layers

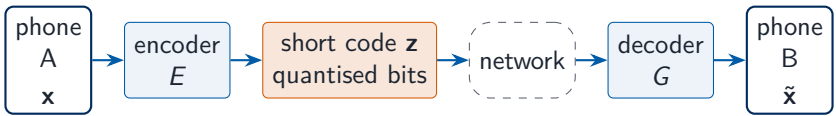


encoder	output	parameters
dense	256	200 960
dense	64	16 448
dense	16	1 040

decoder	output	parameters
dense	64	1 088
dense	256	16 640
dense	784	201 488

437 664 trainable parameters; ReLU after every hidden layer, sigmoid at the output

Compression with Learned Codecs



- JPEG: a fixed transform (discrete cosine transform and quantisation)
- learned image codecs: an autoencoder with a quantised, entropy-coded code (Ballé et al., ICLR 2018)
- speech: the Lyra codec with the learned SoundStream encoder and decoder

The Largest Errors Mix 7s and Badly Written Digits



The 16 test digits with the largest error, labelled by their true digit: 7s (orange labels) and badly written digits of the trained classes.

A Subtle and a Clear Anomaly Compared

anomaly	7s left out of training	photographs
reconstruction error, normal digits	0.013 (mean)	0.011 (median)
reconstruction error, anomalies	0.018 (mean)	0.219 (median)
flagged at 5% false alarms	13%	99.8%
AUC	0.70	0.9997

An anomaly far from the training data is easy to detect; one that resembles it is hard.

8

2D Ising Model: Reference Methods

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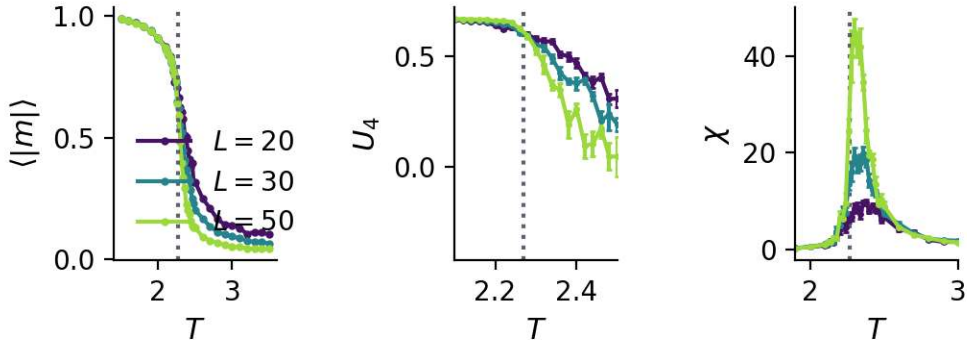
Observables: Definition and Estimator from Samples

$\langle O\rangle = \frac{1}{Z}\sum_{\mathbf{s}} O(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$; Monte Carlo gives n samples $\mathbf{s}^{(k)} \sim p$, $e_k = H(\mathbf{s}^{(k)})/N$, $m_k = \frac{1}{N}\sum_i s_i^{(k)}$, $\overline{x} = \frac{1}{n}\sum_k x_k$

	definition through Z	estimator	measures
energy	$\langle e\rangle = -\frac{1}{N}\partial_\beta \ln Z$	$\overline{e}$	bonds
magnetisation	$\langle\lvert m\rvert\rangle = \frac{1}{Z}\sum_{\mathbf{s}} \lvert m(\mathbf{s})\rvert\, e^{-\beta H}$	$\overline{\lvert m\rvert}$	order
specific heat	$c = \frac{\beta^2}{N}\partial_\beta^2 \ln Z = N\beta^2(\langle e^2\rangle - \langle e\rangle^2)$	$N\beta^2(\overline{e^2} - \overline{e}^2)$	energy fluctuations
susceptibility	$\chi = N\beta(\langle m^2\rangle - \langle\lvert m\rvert\rangle^2)$	$N\beta(\overline{m^2} - \overline{\lvert m\rvert}^2)$	m fluctuations
Binder cumulant	$U_4 = 1 - \langle m^4\rangle/3\langle m^2\rangle^2$	$1 - \overline{m^4}/3\,\overline{m^2}^2$	shape of $P(m)$

$\langle m\rangle = 0$ on a finite lattice ($\mathbf{s} \to -\mathbf{s}$), hence $|m|$; χ with $\langle|m|\rangle$: the finite-lattice form, not $\partial_h \ln Z$; variances divide by n

The Reference: T_c from Finite-Size Scaling

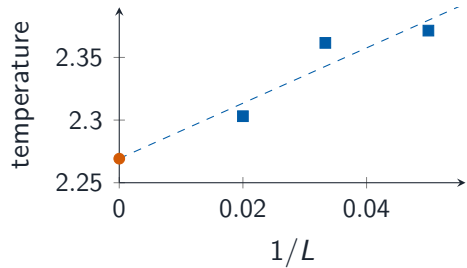


$m = \frac{1}{N}\sum_i s_i, \quad U_4 = 1 - \frac{\langle m^4\rangle}{3\langle m^2\rangle^2}, \quad \chi = \frac{N}{T}(\langle m^2\rangle - \langle|m|\rangle^2);$ lattices $L \times L$, $L = 20, 30, 50$

Binder Crossings and Susceptibility Peaks Give $T_c = 2.269$

We locate T_c from the magnetisation $\langle |m| \rangle$, the Binder cumulant U_4 and the susceptibility χ as functions of T.

- U_4 curves for different L **cross at** T_c: 2.250 ($L = 20/30$), 2.278 ($L = 30/50$)
- the peak of χ moves as $T_c(L) = T_c + a/L$: 2.371, 2.362, 2.303; fit: $T_c = 2.269$, exact 2.2692



squares: peaks of χ for $L = 50, 30, 20$; dashed: linear fit in $1/L$; dot: exact T_c

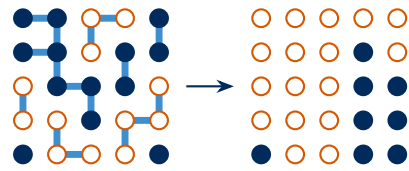
K. Binder, Z. Phys. B **43**, 119 (1981), doi:10.1007/BF01293604; M. E. J. Newman, G. T. Barkema, *Monte Carlo Methods in Statistical Physics*, Oxford University Press (1999)

Sampling near T_c: Flip Whole Clusters

Near T_c single-spin Metropolis flips are rejected inside large domains: successive samples are almost copies (*critical slowing down*).

Swendsen–Wang update:

1. bond each pair of parallel neighbours with probability $p = 1 - e^{-2\beta J}$
2. find the clusters connected by bonds
3. flip every cluster with probability $1/2$



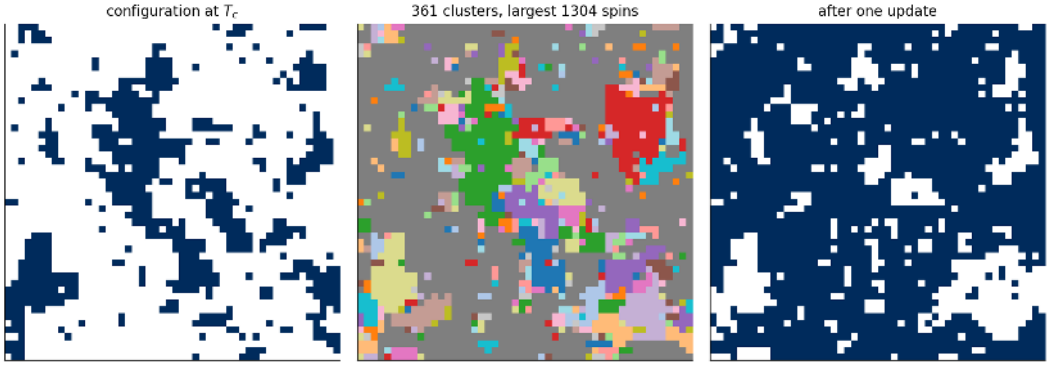
exact: $e^{\beta J s_i s_j} = e^{\beta J}\left[(1-p) + p\,\delta_{s_i s_j}\right]$

$L = 50$ at T_c: autocorrelation time 26.4 Metropolis sweeps vs 3.4 cluster updates

Filled circles $s_i = +1$, open circles $s_i = -1$. Bonds (light blue) join only parallel neighbours.

R. H. Swendsen, J.-S. Wang, Phys. Rev. Lett. **58**, 86 (1987), doi:10.1103/PhysRevLett.58.86

One Swendsen–Wang Update at T_c



left: a configuration at T_c; middle: the clusters connected by bonds, one colour per cluster; right: the configuration after every cluster was flipped with probability $1/2$. On this 50×50 lattice one update forms 361 clusters, and the largest contains 1304 of the 2500 spins.

Below T_c the Configurations Form Two Groups

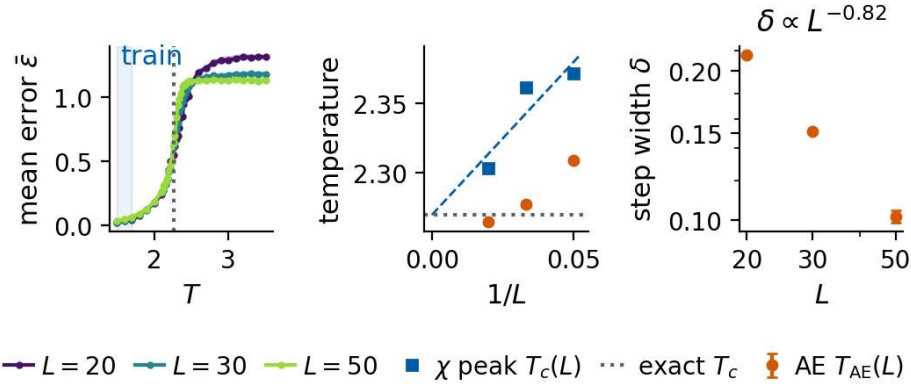
The projections $w^{(l)}$ of the 9472 configurations on the principal directions $\boldsymbol{\lambda}_l$ arrange themselves by temperature.

- low T: two groups at $w^{(1)} \approx \pm 30 = \pm\sqrt{N}$, all spins up or all down; high T: one group around 0
- $\langle |w^{(1)}| \rangle / \sqrt{N}$ lies on the magnetisation $\langle |m| \rangle$ (correlation 0.99996 between $|w^{(1)}|$ and $|m|$); $\boldsymbol{\lambda}_1$ carries 37% of the variance
- the Binder crossings computed from $w^{(1)}$ agree with those from m

Binder crossing	from m	from $w^{(1)}$
$L = 20/30$	2.2495	2.2495
$L = 30/50$	2.2784	2.2773

L. Wang, Phys. Rev. B **94**, 195105 (2016), doi:10.1103/PhysRevB.94.195105; S. J. Wetzel, Phys. Rev. E **96**, 022140 (2017), doi:10.1103/PhysRevE.96.022140

The Autoencoder Signal Sharpens with the System Size



reconstruction error of one configuration $\varepsilon(\mathbf{s}) = \frac{1}{N}\sum_i (s_i - \tilde{s}_i)^2$; $\bar{\varepsilon}(T)$, its mean at T; fit $\bar{\varepsilon}(T) = a + b\tanh\big((T - T_{\mathrm{AE}})/\delta\big)$ on $1.9 \le T \le 3.0$; centre T_{AE} (AE $T^*(L)$ in the legend), width δ (Δ in the right panel)

The Step Centre Approaches T_c and the Step Narrows

We train the autoencoder on the 768 configurations with $T \leq 1.7$ (Yang: $M(1.7) = 0.970$), with Adam for 200 epochs, and fit a step to $\bar{\varepsilon}(T)$. The table lists its centre T_{AE} and width δ, each the mean over three seeds, next to the exact $T_c = 2/\ln(1+\sqrt{2}) = 2.2692$.

L	T_{AE}	$T_{\mathrm{AE}} - T_c$	δ
20	2.3085	$+0.039$	0.215
30	2.2767	$+0.008$	0.151
50	2.2641	-0.005	0.102

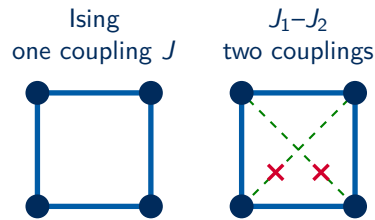
- T_{AE} approaches T_c as L grows; three seeds agree to ± 0.002
- $\delta \propto L^{-0.82}$; the finite-size rounding of the transition scales as $L^{-1/\nu} = L^{-1}$ (three sizes only: indicative)
- the autoencoder is not a thermodynamic observable: no derived finite-size law, so we do not extrapolate T_{AE}

9

J_1–J_2 Model: More Details

Motivation for a Second Model

- for the Ising model we *knew* the answer: one ordered phase, one order parameter m, T_c from Onsager
- in real materials the spins feel **competing interactions**; which order appears, and where, is not known in advance
- **classical model:** $s_i = \pm 1$, each configuration a snapshot of the Boltzmann distribution

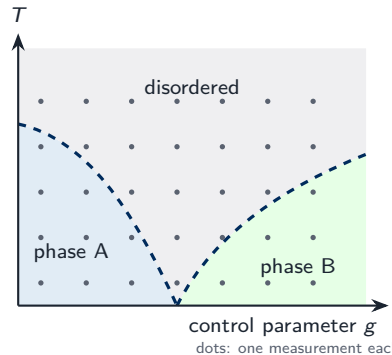


Four parallel spins on a square. The edges ($J_1 > 0$, solid) prefer parallel spins, the diagonals ($J_2 > 0$, dashed) antiparallel ones. No configuration of the four spins satisfies all six bonds.

Mapping an Unknown Phase Diagram Raises Two Problems

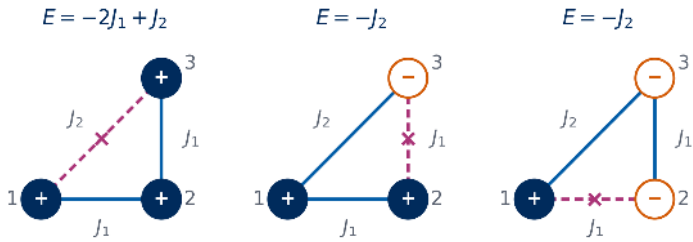
a phase diagram is mapped by measuring on a grid of control parameters:

- **the quantity to plot**: the wrong observable hides a phase
- **a dense grid is expensive**, in the laboratory and on the computer



the boundaries (dashed) are unknown; an observable sensitive only to phase A would show phase B as "disordered"; every grid point costs a measurement

No Configuration Satisfies Every Bond



every state breaks at least one bond (dashed)

- In $H_\triangle = -J_1(s_1 s_2 + s_2 s_3) + J_2\, s_1 s_3$ the J_1 bonds favour $s_1 = s_2 = s_3$ and the J_2 bond favours $s_1 = -s_3$, so every state breaks a bond. This is **frustration**.
- Every loop whose coupling signs multiply to -1, here $(+, +, -)$, is frustrated.
- The lattice settles in the **lowest-energy compromise**, ferromagnet for $g < 1/2$, stripes for $g > 1/2$.

Mean Field Misses the Effect of Frustration

Mean field (each spin feels the average of its neighbours):

$$T_{\text{ferro}}^{\text{MF}} = 4(J_1 - J_2), \qquad T_{\text{stripes}}^{\text{MF}} = 4J_2,$$

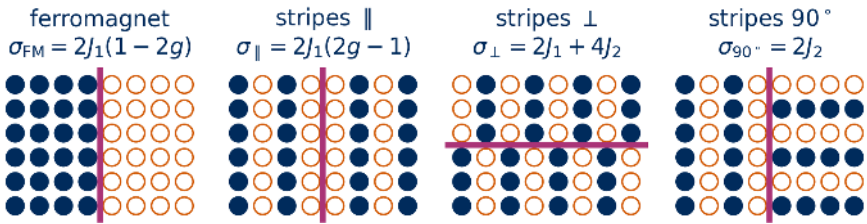
meeting at $g = 1/2,\ T = 2$.

Fluctuations between the degenerate states destroy the order, and the true $T_c \to 0$ at $g = 1/2$; mean field also gives 4 instead of 2.269 at $g = 0$.

g	0	0.3	0.45	0.7	1.0
mean field T^{MF}	4.000	2.800	2.200	2.800	4.000
transfer matrices of strips	2.272	1.266	0.711	1.290	2.078

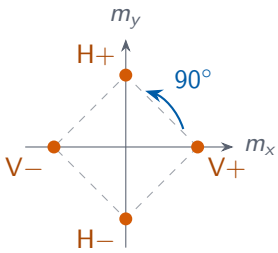
transfer matrices: crossing of ξ_L/L for strip widths 8 and 10

Domain Walls Cost No Energy at $g = 1/2$



- The energy per unit length σ of each wall at $T = 0$ follows from counting the broken bonds. The cheapest walls cost $\sigma_{\min} = 2J_1|1-2g|$.
- Long walls are suppressed while $\sigma_{\min} > T \ln 3$ (a rough Peierls estimate), so the ordering temperature **drops to zero at** $g = 1/2$.
- At $g = 1/2$ whole rows or columns flip at no cost, which gives $2^L + 2^M - 2$ ground states on an $L \times M$ torus.

Stripe Order Breaks Spin Flip and Lattice Rotation



$m_x = m_{(\pi,0)}$, $m_y = m_{(0,\pi)}$

- The ferromagnet has two ordered states, $m = \pm 1$, and breaks only the spin flip.
- The stripes have **four states**, V± and H± (vertical, horizontal), at the corners of a square. Flips, translations and 90° rotations permute them.
- For $g \to \infty$ the sublattices with $x + y$ even and odd decouple, and each orders with its own sign $\sigma_A, \sigma_B = \pm 1$.
- Two coupled Ising variables form the **Ashkin–Teller model**, whose exponents vary along its transition line.

J. Ashkin, E. Teller, Phys. Rev. **64**, 178 (1943), doi:10.1103/PhysRev.64.178; A. Kalz, A. Honecker, M. Moliner, Phys. Rev. B **84**, 174407 (2011), doi:10.1103/PhysRevB.84.174407

Metropolis Sampling of the (g, T) Grid

- Metropolis Monte Carlo, all couplings and chains in parallel, each chain cooled from $T = 4$ to $T = 0.1$
- 32×32 lattice; 81 values of g, 79 temperatures, 100 snapshots per point; the whole grid, 639 900 snapshots, took 1186 s of computing time
- high T: noise; low T: uniform ($g < 1/2$) or stripes ($g > 1/2$)
- near $g = 1/2$: ferromagnetic and stripe patches coexist; at $g = 1/2$, $T = 0.3$ the stripes contain doubled rows, one of the many degenerate ground-state patterns

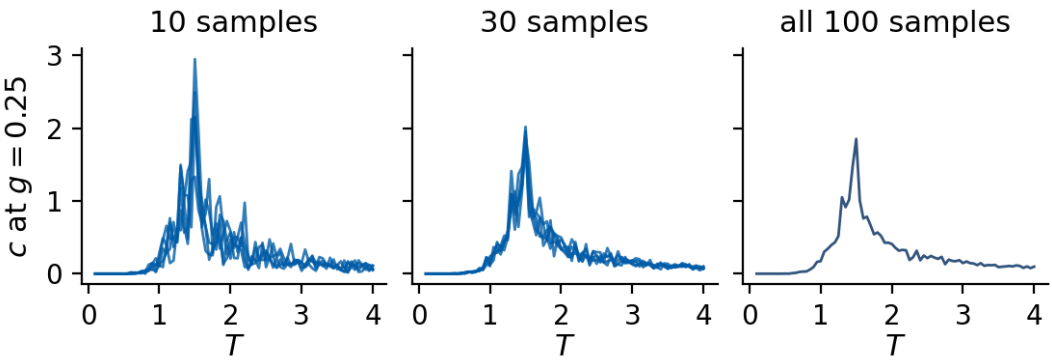
Observables of the J_1–J_2 Model

$\langle O\rangle = \frac{1}{Z}\sum_{\mathbf{s}} O(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$, $Z = \sum_{\mathbf{s}} e^{-\beta H(\mathbf{s})}$, $\beta = 1/T$; n Monte Carlo samples per (g, T), $\overline{x} = \frac{1}{n}\sum_k x_k$; $e = H/N$ with both couplings

	definition through Z	estimator	measures
energy	$\langle e\rangle = -\frac{1}{N}\partial_\beta \ln Z$	$\overline{e}$	bonds
magnetisation	$\langle\lvert m\rvert\rangle = \frac{1}{Z}\sum_{\mathbf{s}} \lvert m(\mathbf{s})\rvert\, e^{-\beta H(\mathbf{s})}$	$\overline{\lvert m\rvert}$	ferromagnetic order
stripe order	$\langle m_{\mathrm{str}}\rangle = \frac{1}{Z}\sum_{\mathbf{s}} m_{\mathrm{str}}(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$	$\overline{m_{\mathrm{str}}}$	stripe order
specific heat	$c = N\beta^2\left(\langle e^2\rangle - \langle e\rangle^2\right)$	$N\beta^2\left(\overline{e^2} - \overline{e}^2\right)$	energy fluctuations

$m_{\mathrm{str}} = \sqrt{m_{(\pi,0)}^2 + m_{(0,\pi)}^2}$ with $m_{(\pi,0)} = \frac{1}{N}\sum_{\mathbf{r}}(-1)^x s_{\mathbf{r}}$ and $m_{(0,\pi)}$ with $(-1)^y$. The spins are multiplied by $+1$ on even and -1 on odd columns, so for perfect vertical stripes all terms have the same sign and $|m_{(\pi,0)}| = 1$. So $m_{\mathrm{str}} = 1$ for perfect stripes of either orientation and $m_{\mathrm{str}} \approx 0$ in the ferromagnet and the paramagnet. The larger of the two gives the stripe orientation.

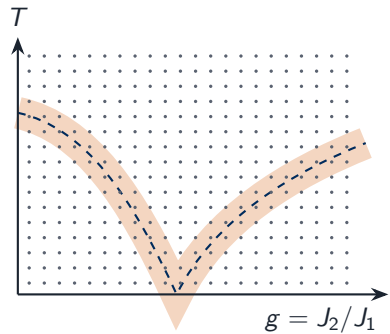
A Dense Scan Is Expensive



- the specific heat at $g = 0.25$ from 10, 30 and all 100 snapshots per temperature; with 10 snapshots the position of the peak jumps between repetitions

A Dense Map Needs 639 900 Configurations

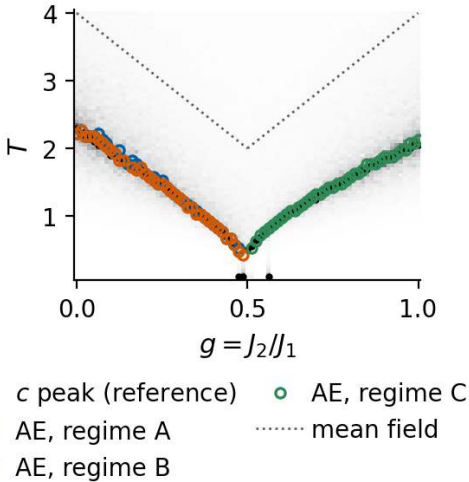
- our map: $81 \times 79 = 6399$ grid points $\times$ 100 snapshots $= 639\,900$ configurations; near a transition consecutive Monte Carlo samples are correlated (critical slowing down), so each costs more
- in a laboratory every grid point is a separate preparation of the sample



Schematic grid with fewer dots than our 6399 points. The boundaries (dashed) occupy a thin band (shaded).

Wanted: a method that needs no order parameter, works from snapshots, and tells us where a dense scan is worth paying for.

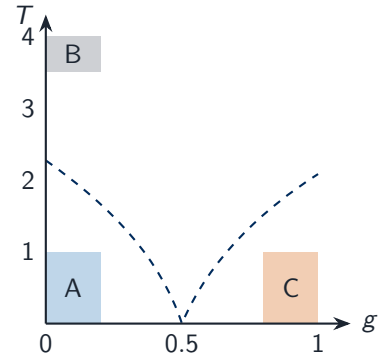
Each Regime Sees the Boundary of Its Own Phase



- boundary point: for every g, the temperature where the mean error changes fastest
- A and B: the ferromagnetic line, $0 \leq g \leq 0.48$ (A), 0.49 (B); C: the stripe line, $0.51 \leq g \leq 1$
- median distance from the specific-heat peak: 0.025, half the temperature step of the grid
- one training region only tells whether a snapshot resembles its training data, so it cannot tell two different anomalies apart

Training Regions A, B and C

- **A, ferromagnet** ($g \leq 0.2,\ T \leq 1$): the ferromagnetic phase is rebuilt almost exactly; stripes and paramagnet are both "not ferromagnet"
- **B, paramagnet** ($g \leq 0.2,\ T \geq 3.5$): it learns little more than the average; the ferromagnet is rebuilt *better* than its own data, the stripe phase is **invisible**
- **C, stripes** ($g \geq 0.8,\ T \leq 1$): the stripe phase and its boundary



training regions A, B, C in the (g, T) plane; dashed: sketch of the transition line

Unassigned Points Lie on the Ordered Side of the Transitions

- **unassigned** (21% of the grid): mostly the ordered side of each transition, where fluctuations make snapshots atypical for a model trained deep in the phase
- around $g = 1/2$ at low T: *paramagnet*, consistent with $T_c \to 0$ at $g = 1/2$: no order within our temperature range

assigned to	A, ferromagnet	B, paramagnet	C, stripes	unassigned
fraction of the grid	9.9%	59.2%	10.1%	20.7%

10

XY Model: More Details

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Configurations and Magnetisation

- spin i: a unit vector in the plane, $\mathbf{s}_i = (\cos\theta_i, \sin\theta_i)$, angle $\theta_i \in [0, 2\pi)$; a configuration is the set of all angles $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_N)$, with energy $H(\boldsymbol{\theta}) = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j)$
- **magnetisation** of one configuration, a two-component vector $\mathbf{m} = (m_x, m_y)$:

$$m_x(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \cos\theta_i, \qquad m_y(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \sin\theta_i$$

- its length depends only on angle differences; its direction turns by α when all angles turn by α:

$$|\mathbf{m}|^2 = m_x^2 + m_y^2 = \frac{1}{N^2} \sum_{i,j} \cos(\theta_i - \theta_j), \qquad 0 \le |\mathbf{m}| \le 1$$

- all angles equal: $|\mathbf{m}| = 1$; independent uniform random angles: $\langle |\mathbf{m}|^2 \rangle = 1/N$

Thermal Averages and the Mean Magnetisation

- **thermal average** at temperature T: the angles are continuous, so the sum over configurations is an integral,

$$\langle O \rangle = \frac{1}{Z} \int_0^{2\pi} d\theta_1 \cdots \int_0^{2\pi} d\theta_N \; O(\boldsymbol{\theta})\, e^{-\beta H(\boldsymbol{\theta})}, \qquad Z = \int d\boldsymbol{\theta}\; e^{-\beta H(\boldsymbol{\theta})}, \quad \beta = 1/T$$

- Monte Carlo draws configurations $\boldsymbol{\theta}^{(k)}$ with probability density $e^{-\beta H}/Z$ and replaces the integral by the sample mean, $\langle O \rangle \approx \frac{1}{n} \sum_{k=1}^{n} O(\boldsymbol{\theta}^{(k)})$

- a global rotation $\theta_i \to \theta_i + \alpha$ leaves H unchanged and rotates $\mathbf{m}$ by α, so $\langle \mathbf{m} \rangle = 0$ on every finite lattice

- we therefore measure $\langle |\mathbf{m}| \rangle$ (or $\langle |\mathbf{m}|^2 \rangle$): nonzero on a finite lattice, $\to 0$ as $L \to \infty$ at every $T > 0$ (Mermin–Wagner)

- example, $T = 0.3$: $\langle |\mathbf{m}| \rangle = 0.940,\ 0.907,\ 0.874$ for $L = 8,\ 32,\ 128$

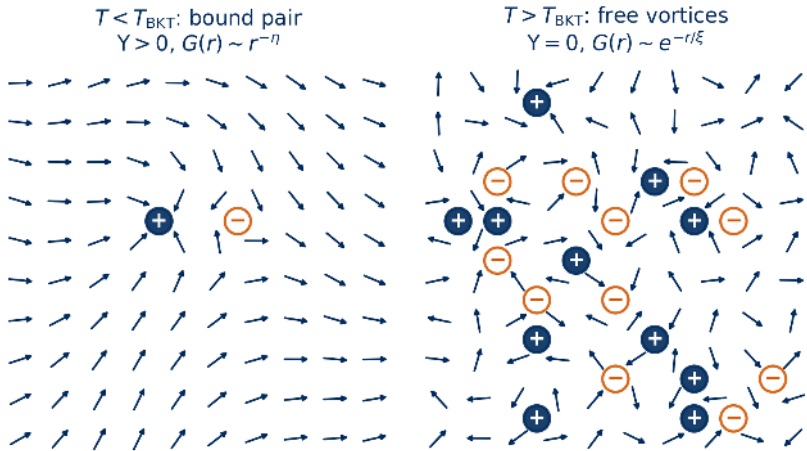
A Continuous Symmetry, Realised in Films and Magnets

The phase of a superfluid or superconducting film and the direction of a planar magnet are angles, exactly like θ_i.

- **symmetry:** H is unchanged by a global rotation $\theta_i \to \theta_i + \alpha$ for every angle α, a continuous symmetry (with the reflection $\theta_i \to -\theta_i$: the group $O(2)$)
- **ground states:** all spins parallel, $E_0 = -2JN$; a continuous family labelled by one angle (Ising: two ground states)
- **realisations:** ^{4}He films (superfluid density measured by Bishop and Reppy, 1978), thin superconducting films, planar magnets
- **Nobel Prize in Physics 2016** (Thouless; Haldane, Kosterlitz): "for theoretical discoveries of topological phase transitions and topological phases of matter"

D. J. Bishop, J. D. Reppy, Phys. Rev. Lett. **40**, 1727 (1978), doi:10.1103/PhysRevLett.40.1727;
nobelprize.org/prizes/physics/2016/summary

Two Phases without a Local Order Parameter



low T: neighbours nearly aligned, the direction drifts slowly, vortices only in bound pairs; high T: directions random beyond a few spacings, free vortices ($+$, $-$: winding ± 1)

Stiffness and Correlations Replace the Order Parameter

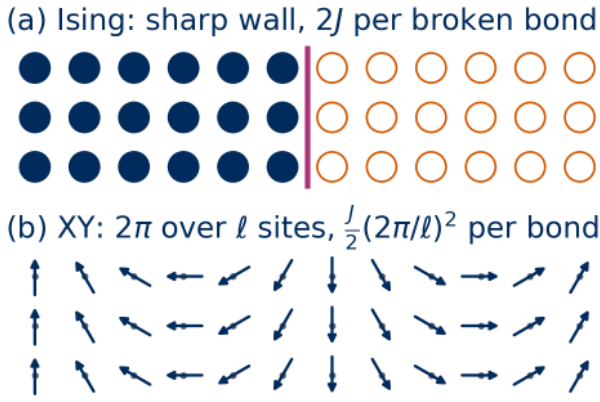
	$T < T_{\mathrm{BKT}}$: quasi-long-range order	$T > T_{\mathrm{BKT}}$: disorder
correlations $G(r)$	power law $r^{-\eta(T)}$	exponential $e^{-r/\xi}$
helicity modulus Υ	$\Upsilon > 0$	$\Upsilon = 0$
vortices	bound in pairs	free
magnetisation, $L \to \infty$	0	0

$G(r) = \langle \cos(\theta_0 - \theta_r) \rangle$: correlation at distance r; ξ: correlation length; Υ: free-energy cost of a twist Δ, $F(\Delta) - F(0) = \frac{\Upsilon}{2}\Delta^2$

- no broken symmetry (Mermin–Wagner); $T_{\mathrm{BKT}} = 0.8929\,J$
- Berezinskii–Kosterlitz–Thouless (BKT) transition, **infinite order**: $\xi \propto e^{b/\sqrt{t}}$, $t = T - T_{\mathrm{BKT}}$, b: a non-universal constant; jump $\Upsilon(T_{\mathrm{BKT}}^{-}) = 2\,T_{\mathrm{BKT}}/\pi$
- 2D Ising, second order: $\xi \propto |t|^{-\nu}$, $\nu = 1$

M. Hasenbusch, J. Phys. A **38**, 5869 (2005), doi:10.1088/0305-4470/38/26/003

A Slow Twist Costs Almost No Energy



turning the spins by 2π over ℓ sites costs $\frac{J}{2}(2\pi/\ell)^2$ per bond, about ℓ^{d-2} in total: independent of ℓ in $d = 2$; an Ising domain wall costs $\propto \ell^{d-1}$

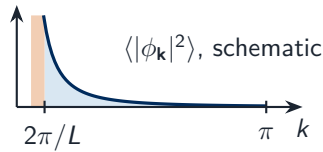
Spin Waves Destroy the Magnetisation

Expanding $\cos x \approx 1 - x^2/2$ gives independent Gaussian modes (spin waves):

$$\langle \phi^2 \rangle \simeq \frac{T}{2\pi J} \ln L, \qquad \langle |\mathbf{m}| \rangle \propto L^{-T/(4\pi J)} \to 0, \qquad G(r) \propto r^{-\eta}, \quad \eta = \frac{T}{2\pi J}$$

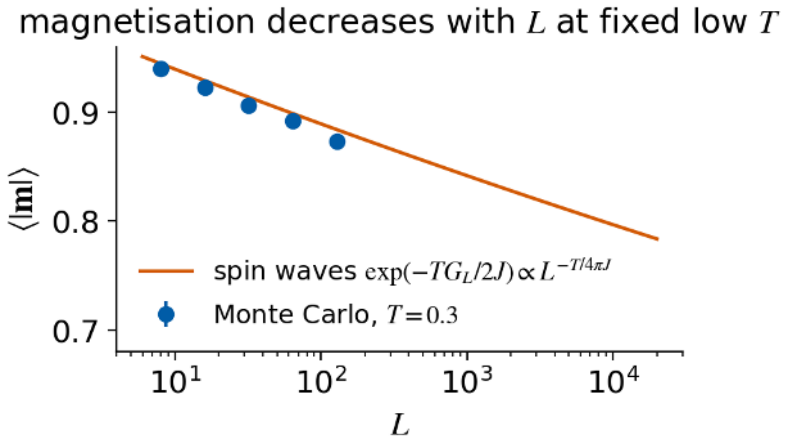
$\phi_i = \theta_i - \bar{\theta}$: angle from the mean direction; $\mathbf{m} = \frac{1}{N} \sum_i (\cos\theta_i, \sin\theta_i)$

- mode $\mathbf{k}$ costs $\frac{J}{2} k^2 |\phi_{\mathbf{k}}|^2$, so equipartition gives $\langle |\phi_{\mathbf{k}}|^2 \rangle = T/(Jk^2)$
- $\langle \phi^2 \rangle = \frac{1}{N} \sum_{\mathbf{k}} \frac{T}{Jk^2} \simeq \frac{T}{2\pi J} \int_{2\pi/L}^{\pi} \frac{dk}{k}$ grows as $\ln L$, carried by the long waves near $k = 2\pi/L$
  - no long-range order at $T > 0$ (Mermin–Wagner)
  - power-law $G(r)$: **quasi-long-range order**
  - **finite-size magnetisation** on every lattice



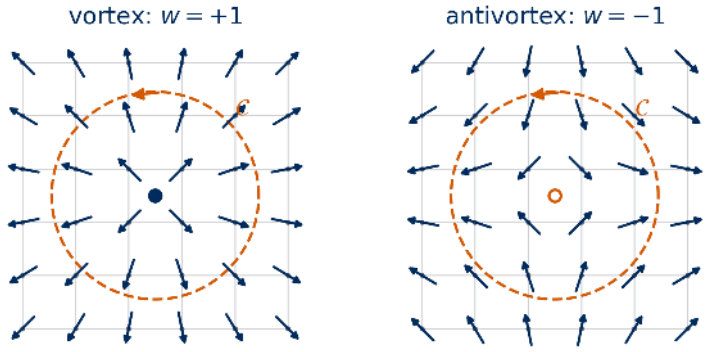
N. D. Mermin, H. Wagner, Phys. Rev. Lett. **17**, 1133 (1966), doi:10.1103/PhysRevLett.17.1133

The Magnetisation Decays Slowly with the Lattice Size



$T = 0.3$: fitted exponent 0.0261 (theory 0.0239); halving $\langle|\mathbf{m}|\rangle$ needs $L \times 3.5 \cdot 10^{11}$

Vortices Are Topological Defects



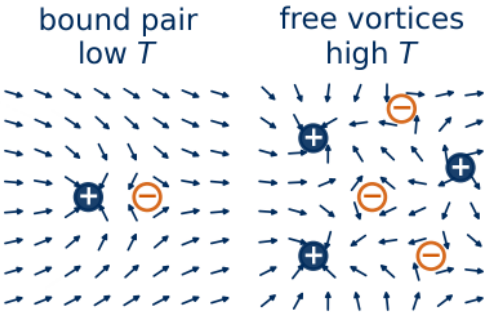
$w = \frac{1}{2\pi}\oint_C \nabla\theta \cdot d\mathbf{l}$ = number of turns of the spins along C

lattice: $w = \frac{1}{2\pi}\sum_{\square}[\theta_j - \theta_i] \in \mathbb{Z}$, the four bonds of a plaquette counter-clockwise, $[x]$: angle difference mapped to $[-\pi, \pi)$, so $w \in \{-1, 0, 1\}$ (-2 has probability zero); on a torus $\sum w = 0$

Entropy Unbinds the Vortices

Because $|\nabla\theta| = 1/r$ around a vortex, its energy grows with the size; its centre can sit on any of $(L/a)^2$ plaquettes:

$$E_v = \pi J \ln \frac{L}{a}, \qquad S_v = 2 \ln \frac{L}{a}, \qquad F_v = E_v - TS_v = (\pi J - 2T) \ln \frac{L}{a}$$



a: lattice spacing

- $T < \pi J/2$: only bound pairs; above: free vortices
- $T_{\mathrm{BKT}} = 0.893$ lies below $\pi J/2 = 1.571$ because bound pairs **screen** the interaction
- vortex density at $T = 0.90$: 0.009, smooth in T

J. M. Kosterlitz, D. J. Thouless, J. Phys. C **6**, 1181 (1973), doi:10.1088/0022-3719/6/7/010

Bound Vortex Pairs Are Present at Every Temperature

Far from a vortex–antivortex pair at distance d the two windings cancel, so the energy of the pair does not grow with L,

$$E_{\mathrm{pair}}(d) \approx 2\pi J \ln \frac{d}{a} + 2E_{\mathrm{core}}.$$

Here a is the lattice spacing and E_{core} the energy of one vortex core.

- On the lattice the slope $dE_{\mathrm{pair}}/d\ln d$ is 6.40 for $d = 8 \to 16$ and 6.31 for $d = 16 \to 32$, close to $2\pi J = 6.28$.
- A pair has a finite energy, so pairs are thermally activated. The vortex density follows $n_v \propto e^{-E_a/T}$ with a fitted $E_a \approx 7.3\,J$ ($T \leq 0.75$, $L = 48$).

The low-temperature phase contains bound pairs at every $T > 0$.

Screening by Bound Pairs Sets the Universal Jump

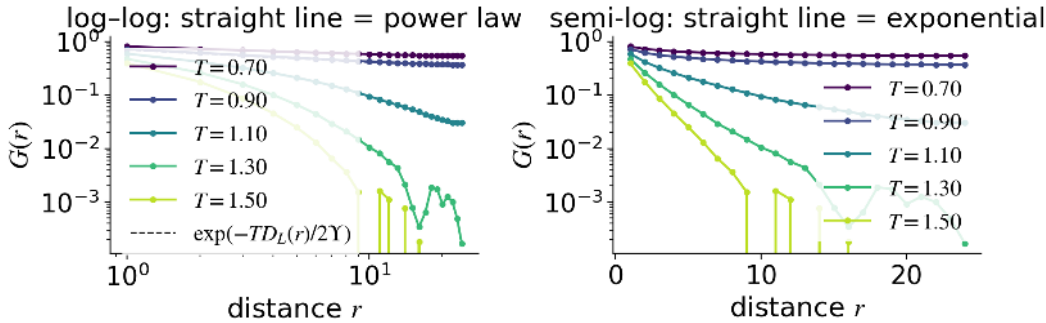
By analogy with electrostatics, vortices act like 2D charges with a logarithmic interaction, and bound pairs act like dipoles in a dielectric that **screen** it. The coupling J in the vortex free energy becomes the stiffness $\Upsilon < J$,

$$F_v = (\pi\Upsilon - 2T)\ln\frac{L}{a}.$$

- Free vortices appear once Υ falls to $2T/\pi$. This condition gives the **universal jump** $\Upsilon(T_{\mathrm{BKT}}^{-}) = 2T_{\mathrm{BKT}}/\pi$ and lowers the transition from $\pi J/2 = 1.571$ to $T_{\mathrm{BKT}} = 0.893$.
- Monte Carlo at $T = 0.70$, $L = 48$ gives $\Upsilon = 0.777$, below the spin-wave value $J - T/4 = 0.825$.

J. M. Kosterlitz, J. Phys. C **7**, 1046 (1974), doi:10.1088/0022-3719/7/6/005; D. R. Nelson, J. M. Kosterlitz, Phys. Rev. Lett. **39**, 1201 (1977), doi:10.1103/PhysRevLett.39.1201

Power-Law Correlations below, Exponential above

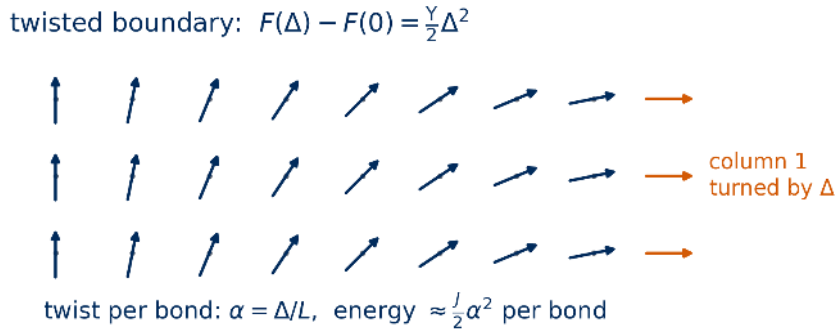


$G(r) = \frac{1}{N}\sum_i \cos(\theta_i - \theta_{i+r})$, $L = 48$

- below: $r^{-\eta}$, $\eta = T/(2\pi\Upsilon)$, spin waves within 1% for $T \leq 0.8$; $\eta(T_{\mathrm{BKT}}) = 1/4$
- above: $e^{-r/\xi}$, $\xi = 9.3$ at $T = 1.05$, 1.5 at $T = 1.50$

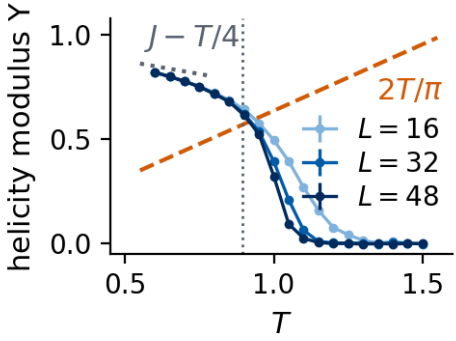
J. M. Kosterlitz, J. Phys. C **7**, 1046 (1974), doi:10.1088/0022-3719/7/6/005

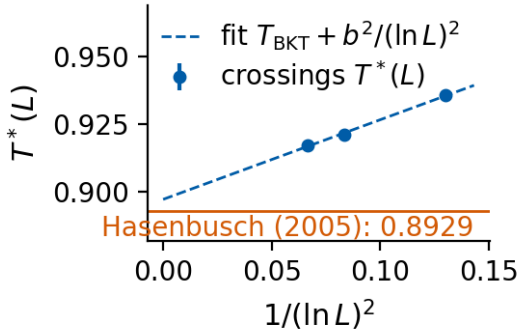
The Stiffness Is the Response to a Twist



A phase with order resists a twist (by analogy, a superfluid resists a phase gradient): $\Upsilon = \frac{1}{N}\partial_\alpha^2 F|_{\alpha=0}$, $\alpha = \Delta/L$ the twist per bond, $F = -T \ln Z$; measured in equilibrium from bond fluctuations

The Stiffness Jump Locates T_{BKT}



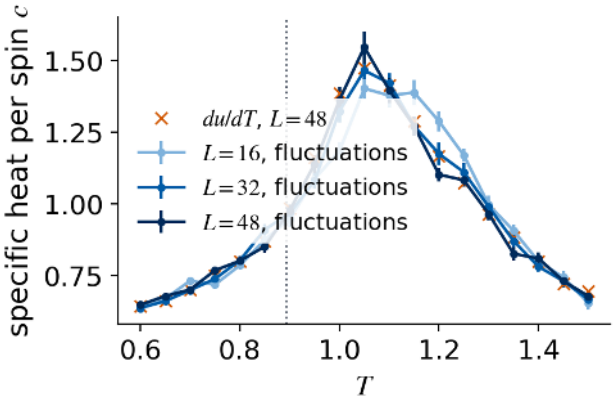


- universal jump $\Upsilon(T_{\mathrm{BKT}}^{-}) = 2T_{\mathrm{BKT}}/\pi$ (Nelson–Kosterlitz); left: $\Upsilon_L(T)$ and $2T/\pi$
- crossings $T^*(L) = 0.936,\ 0.921,\ 0.917$ ($L = 16, 32, 48$); $\xi(T^*) \approx L$ gives $T^*(L) = T_{\mathrm{BKT}} + b^2/(\ln L)^2$; fit (right): 0.897 ± 0.003, literature 0.8929

D. R. Nelson, J. M. Kosterlitz, Phys. Rev. Lett. **39**, 1201 (1977), doi:10.1103/PhysRevLett.39.1201

The Specific Heat Peaks above T_{BKT}

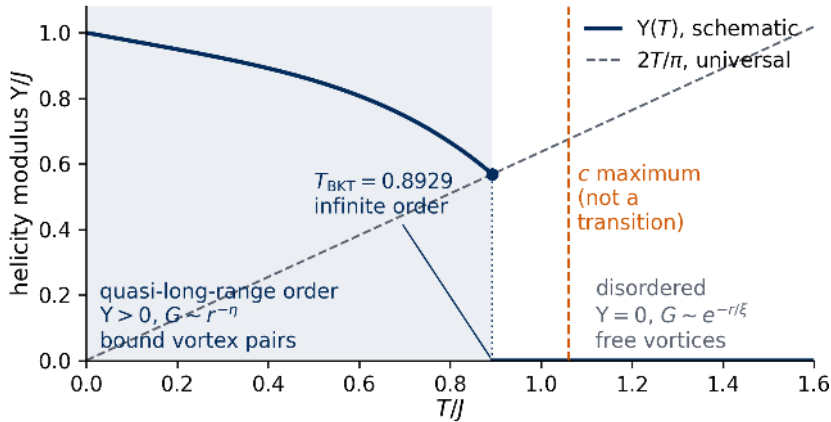
$c = (\langle H^2 \rangle - \langle H \rangle^2)/(NT^2)$: specific heat per spin



dotted: T_{BKT}

- singular free energy $\propto \xi^{-2} \propto e^{-2b/\sqrt{t}}$: all derivatives vanish at $t \to 0^+$; no divergence
- maximum at $T = 1.12,\ 1.07,\ 1.06$ $(L = 16, 32, 48)$: **not a transition**
- **warning:** a detector based on the energy finds this maximum

The Phase Diagram on the Temperature Axis



schematic $\Upsilon(T)$; T_{BKT}: infinite-order transition; the c maximum lies in the disordered phase

Reference Temperatures for the Comparison

Every temperature an unsupervised method returns is compared with these values.

reference	value	meaning
T_{BKT} (Monte Carlo, Hasenbusch)	0.8929	the transition
stiffness crossings $T^*(L)$	0.936, 0.921, 0.917 $\to$ 0.897(3)	estimates of T_{BKT}
specific-heat maximum	1.12, 1.07, 1.06	not a transition
estimate $\pi J/2$	1.571	no screening

values for $L = 16, 32, 48$; data for the machine learning: Monte Carlo snapshots, $L = 16, 32, 48$, 19 temperatures 0.60–1.50, 192 snapshots per temperature, no labels

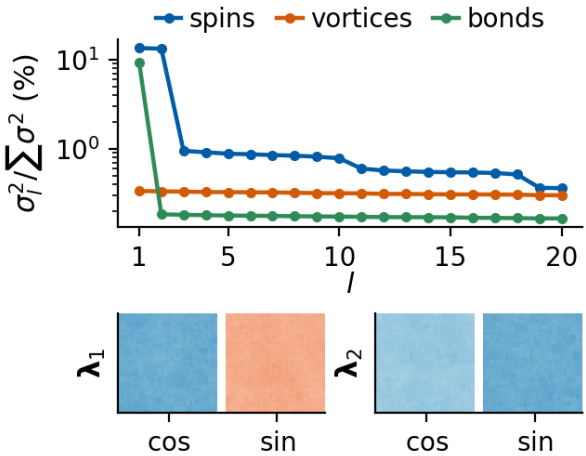
Thermodynamic Observables: Definition and Estimator

$Z = \int d\boldsymbol{\theta}\, e^{-\beta H}$, $\beta = 1/T$, $\langle O \rangle = Z^{-1} \int d\boldsymbol{\theta}\, O\, e^{-\beta H}$; estimator: sample mean $\overline{x} = \frac{1}{n} \sum_k x_k$ over $n = 24$ chains $\times$ 1000 Monte Carlo steps per T

	definition through Z	**estimator**
energy	$u = -\frac{1}{N} \partial_\beta \ln Z$	$\overline{E}/N$
specific heat	$c = \frac{\beta^2}{N} \partial_\beta^2 \ln Z$	$\frac{\beta^2}{N} (\overline{E^2} - \overline{E}^2)$
magnetisation	$\langle \lvert \mathbf{m} \rvert \rangle$	$\overline{\lvert \mathbf{m} \rvert}$
helicity modulus	$\Upsilon = \frac{1}{N} \partial_\alpha^2 F \lvert_0$, $F = -T \ln Z(\alpha)$	$\frac{1}{2N} [\overline{C_x + C_y} - \beta\, \overline{S_x^2 + S_y^2}]$
vortex density	$\langle n_v \rangle$	$\overline{n_v}$

E: energy of a step; $C_\mu = \sum_r \cos \Delta_r^\mu$, $S_\mu = \sum_r \sin \Delta_r^\mu$, $\Delta_r^\mu = \theta_{r+\hat{\mu}} - \theta_r$; α: twist added to every Δ_r^x; n_v: fraction of plaquettes with $w \neq 0$; variances divide by n

PCA Spectra of the Three Representations

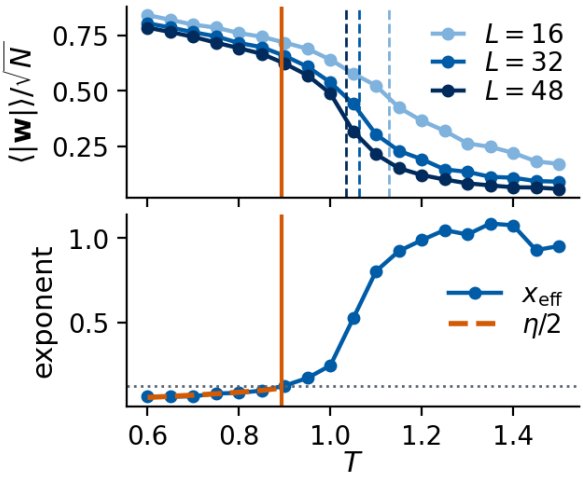


Principal component analysis (PCA) finds the directions $\boldsymbol{\lambda}_l$ of largest variance, with variance fractions $\sigma_l^2/\sum\sigma^2$. We use all snapshots of $L = 32$.

- **spins:** $\boldsymbol{\lambda}_{1,2}$ carry 13.4, 13.2%, uniform: (m_x, m_y)
- **bonds:** 9.3%: energy
- **vortices:** flat, no linear pattern

W. Hu, R. R. P. Singh, R. T. Scalettar, Phys. Rev. E **95**, 062122 (2017), doi:10.1103/PhysRevE.95.062122

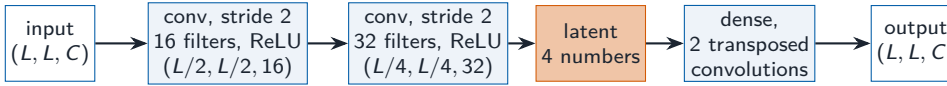
PCA Finds the Finite-Lattice Magnetisation



The top panel shows $\langle|\mathbf{w}|\rangle/\sqrt{N} = \langle|\mathbf{m}|\rangle$, with $\mathbf{w} = (w^{(1)}, w^{(2)})$ and $w^{(l)} = \boldsymbol{\lambda}_l^{\top}(\mathbf{x} - \bar{\mathbf{x}})$, unrelated to the vortex winding w_r. The bottom panel shows the exponent x_{eff} in $\langle|\mathbf{w}|\rangle \propto L^{-x_{\text{eff}}}$ together with $\eta/2$.

- the finite-size magnetisation, $\langle|\mathbf{m}|\rangle \propto L^{-\eta/2}$
- steepest fall $T_{\text{PCA}} = 1.13,\ 1.06,\ 1.03$ $(L = 16, 32, 48)$: **drifts with L**

Convolutional Autoencoder on the Torus

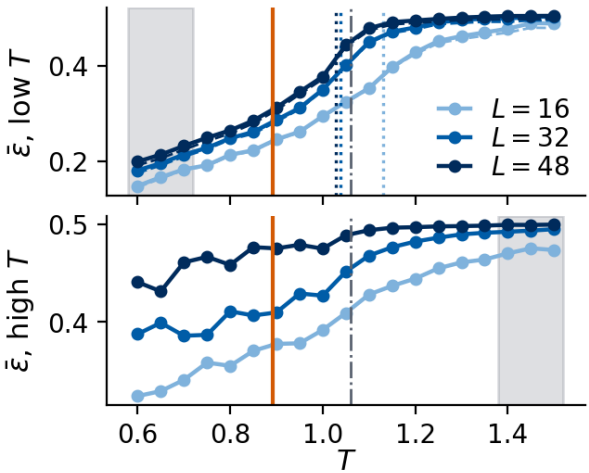


An autoencoder (AE) compresses a snapshot to a few numbers and reconstructs it; trained on one well-understood phase, it measures at every T how much the snapshots differ from that phase.

- periodic padding (the lattice is a torus); C input channels; ReLU $= \max(0, x)$
- **training:** $T \leq 0.70$, 288 snapshots of chains 0–11, random rotations; Adam optimiser, 40 epochs, two seeds
- **reconstruction error** $\varepsilon(\mathbf{x}) = \frac{1}{NC} \sum_{r,c} (x_{r,c} - \tilde{x}_{r,c})^2$; $\bar{\varepsilon}(T)$: mean over 96 held-out snapshots (chains 12–23)
- **estimate** $T_{\mathrm{AE}}(L)$: steepest rise of $\bar{\varepsilon}(T)$; error: bootstrap over chains

K. Kottmann, P. Huembeli, M. Lewenstein, A. Acín, Phys. Rev. Lett. **125**, 170603 (2020), doi:10.1103/PhysRevLett.125.170603

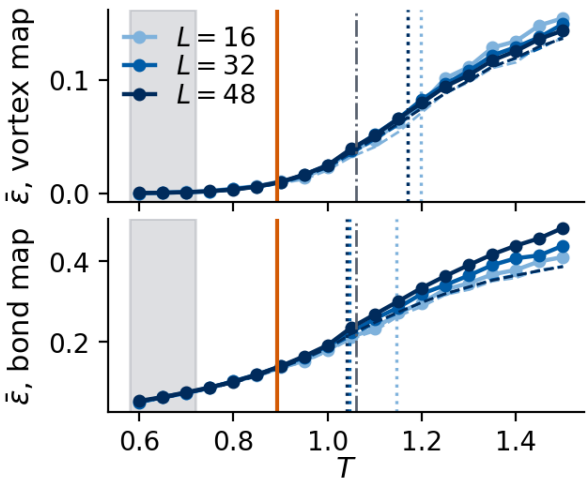
On Raw Spins the Autoencoder Learns the Magnetisation



dashed: $(1 - |\mathbf{m}|^2)/2$, error of $\tilde{\mathbf{s}}_r = \mathbf{m}$; orange: T_{BKT}; grey: training T

- **low-T training:** error = magnetisation baseline; $T_{\mathrm{AE}} = 1.13,\ 1.04,\ 1.03$
- **high-T training:** ordered snapshots look *less* anomalous

The Input Representation Decides the Signal



dashed: $\frac{1}{N}\sum_r w_r^2 \approx n_v$ (top), variance of b_r (bottom)

- **vortices:** error $= n_v$; $T_{\mathrm{AE}} = 1.20,\ 1.17,\ 1.17,$ **above the c maximum**
- **bonds:** $T_{\mathrm{AE}} = 1.15,\ 1.05,\ 1.04$: the c maximum
- counts defects, not whether they are bound

Larger Networks Give the Same Temperature

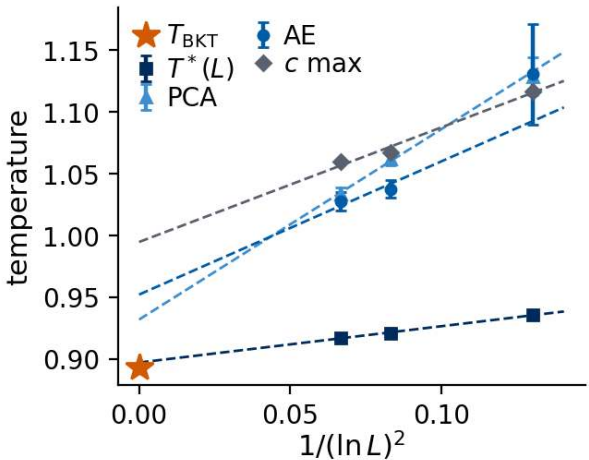
autoencoder on raw spins, $L = 32$, one change at a time, two seeds each; final training loss and T_{AE} bootstrap error

	baseline	latent 2	latent 8	latent 16	width $\times 2$	width $\times 4$	epochs $\times 3$
loss	0.194	0.209	0.187	0.173	0.192	0.189	0.184
T_{AE}	1.038(7)	1.037(7)	1.037(8)	1.039(8)	1.036(7)	1.038(7)	1.038(7)

baseline: 4 latent numbers, 16 and 32 filters, 40 epochs; width $\times k$: all filter numbers multiplied by k

- the loss changes, the temperature does not: largest shift 0.14 bootstrap errors (vortex AE: 0.10)
- **within these variants, capacity and training time do not move T_{AE}**

The Magnetisation Crossover Extrapolates towards T_{BKT}



fit $T(L) = T_\infty + b^2/(\ln L)^2$, from $\xi(T) \approx L$

- T^*: 0.897 ± 0.003; PCA: 0.932 ± 0.020; AE: 0.952 ± 0.036
- $T_\infty = 0.8929$ fixed: $p = 0.12,\ 0.13$: T_{BKT} not excluded
- c max: 0.995
- **3 sizes; the form is BKT physics**

Snapshots Reveal a Crossover above T_{BKT}

method	measures	T, $L = 48$	vs. the known physics
stiffness crossing	$\Upsilon_L = 2T/\pi$	0.917	$\to 0.897(3)$, $T_{\mathrm{BKT}} = 0.8929$
specific heat	maximum of c	1.06	smooth, above T_{BKT}
PCA on spins	finite-size $\lvert\mathbf{m}\rvert$	1.03	crossover $\to 0.932 \pm 0.020$
AE on spins	finite-size $\lvert\mathbf{m}\rvert$	1.03	crossover $\to 0.952 \pm 0.036$
AE on bonds	energy fluctuations	1.04	specific-heat maximum
AE on vortices	vortex density	1.17	smooth proliferation

An unsupervised detector finds what varies most in the snapshots; the physics tells us what that is and how to extrapolate it.

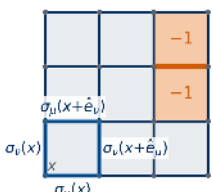
11

Order without a Local Order Parameter



JAGIELLONIAN UNIVERSITY
IN KRAKOW

Ising Lattice Gauge Theory: Variables on the Links



$$H = -J\sum_p U_p$$

$$U_p = \sigma_\mu(x)\,\sigma_\nu(x+\hat{e}_\mu)\,\sigma_\mu(x+\hat{e}_\nu)\,\sigma_\nu(x)$$

Ising variables $\sigma = \pm 1$ live on the links, not on the sites

link $\sigma = +1$

link $\sigma = -1$

plaquette $U_p = +1$: energy $-J$

plaquette $U_p = -1$: energy $+J$

- links $\sigma_\mu(x) = \pm 1$ (unrelated to the PCA variances σ_l^2); coupling $\beta = J/T$
- local symmetry: $\sigma_\mu(x) \to \gamma(x)\,\sigma_\mu(x)\,\gamma(x+\hat{e}_\mu)$, $\gamma(x) = \pm 1$, leaves every U_p unchanged
- Elitzur: a local symmetry cannot break, so $\langle \sigma_\mu(x) \rangle = 0$ in every phase

F. J. Wegner, J. Math. Phys. **12**, 2259 (1971), doi:10.1063/1.1665530; S. Elitzur, Phys. Rev. D **12**, 3978 (1975), doi:10.1103/PhysRevD.12.3978

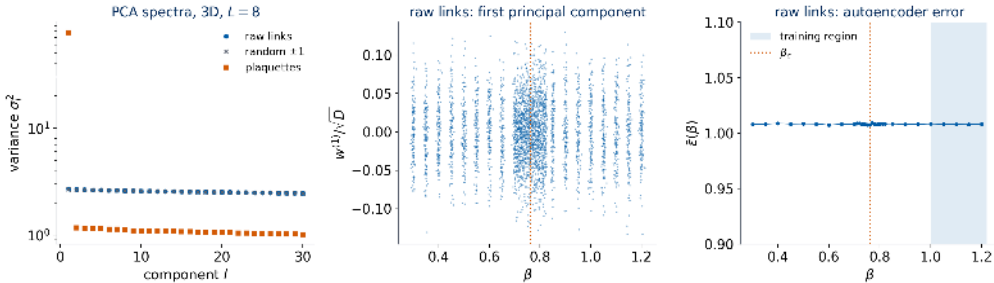
Wilson Loops Distinguish the Phases

Wilson loop $W(R, S)$: thermal average of the product of the links around an $R \times S$ rectangle

- confined: area law $W \sim e^{-\sigma_s RS}$, string tension $\sigma_s > 0$
- deconfined: perimeter law $W \sim e^{-2\kappa(R+S)}$, constant $\kappa > 0$
- 3D: continuous transition at $\beta_c = 0.7614$ (Wegner duality); 2D: no transition, $\langle U_p \rangle = \tanh \beta$

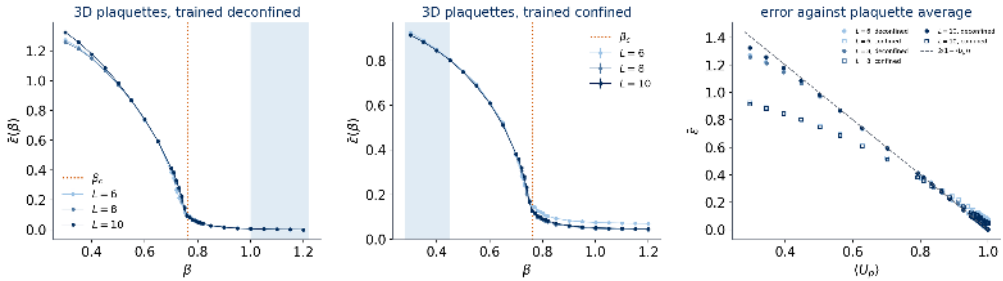
K. G. Wilson, Phys. Rev. D **10**, 2445 (1974), doi:10.1103/PhysRevD.10.2445

Raw Links Look Like Random Signs



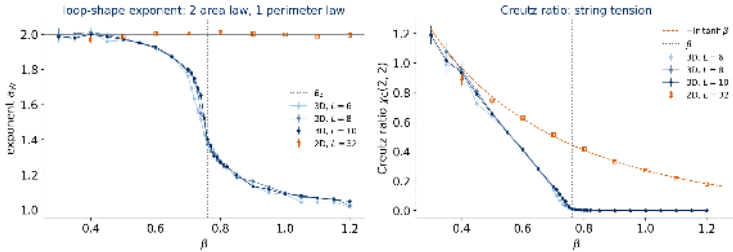
- 3D, $L = 8$, every configuration in a random gauge; left: PCA variances, raw links against random ± 1 and plaquettes
- middle: first PCA coordinate against β; right: autoencoder error, 1.0075 to 1.0093 at every β

Gauge-Invariant Plaquettes Give the Energy



- autoencoder on the plaquettes U_p, trained deconfined ($\beta \geq 1.0$) or confined ($\beta \leq 0.45$); dotted: β_c
- steepest change $\beta^* = 0.715, 0.735, 0.745$ ($L = 6, 8, 10$), at the specific-heat peak
- right: errors against $\langle U_p \rangle$ collapse; the detector measures the energy

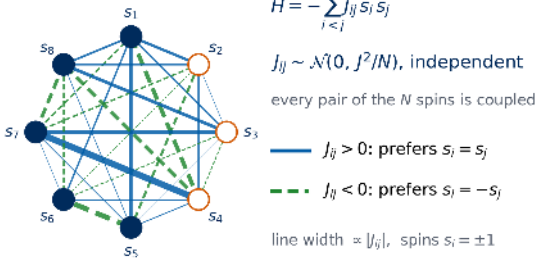
Wilson-Loop Combinations Encode Confinement



- left: $a_W = \log_2[\ln W(2,2)/\ln W(1,1)]$ (labelled a): 2 for an area law, 1 for a perimeter law; passes 1.5 at $\beta = 0.742, 0.751, 0.752$
- right: Creutz ratio $\chi_C(2,2) = -\ln[W(2,2)W(1,1)/W(2,1)^2]$ (labelled $\chi(2,2)$), the string tension
- these combinations come from physics; no unsupervised method found them

M. Creutz, Phys. Rev. D **21**, 2308 (1980), doi:10.1103/PhysRevD.21.2308

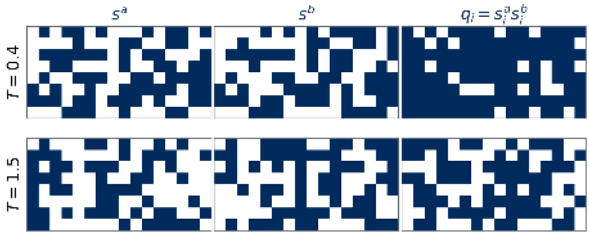
The Sherrington–Kirkpatrick Spin Glass



- Sherrington–Kirkpatrick (SK) model: random couplings drawn once and kept fixed (one disorder sample); $J = 1$
- below $T_c = J$ the spins freeze in random directions; the magnetisation stays zero
- $[\cdot]$: average over disorder samples; $\langle\cdot\rangle$: thermal average in one sample

D. Sherrington, S. Kirkpatrick, Phys. Rev. Lett. **35**, 1792 (1975), doi:10.1103/PhysRevLett.35.1792

The Order Shows in the Overlap of Two Replicas

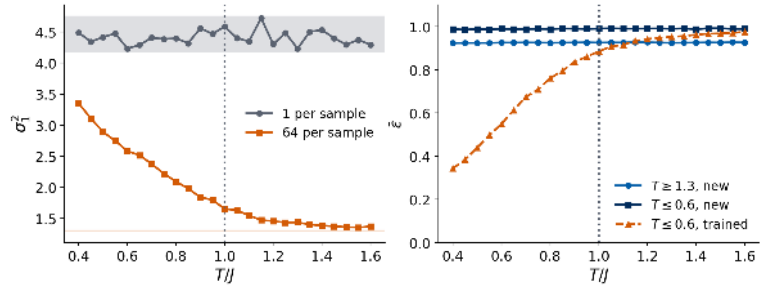


two replicas $\mathbf{s}^a$, $\mathbf{s}^b$ (same couplings) and their overlap field $q_i = s_i^a s_i^b$

- overlap $q = \frac{1}{N}\sum_i s_i^a s_i^b$; below T_c its distribution $P(q)$ has two peaks
- Binder cumulant $g_{\mathrm{B}} = \frac{1}{2}(3 - [\langle q^4\rangle]/[\langle q^2\rangle]^2)$
- crossings $0.870, 0.920, 0.894$, below $T_c = J$

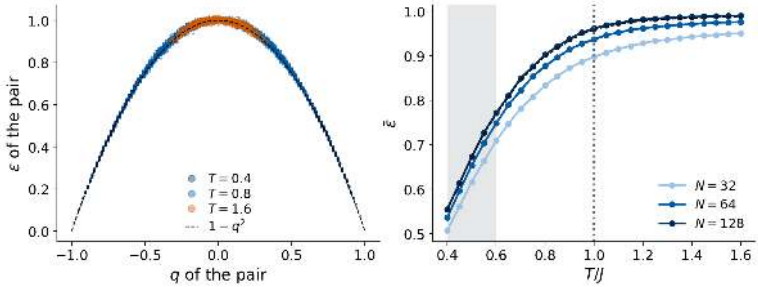
G. Parisi, Phys. Rev. Lett. **43**, 1754 (1979), doi:10.1103/PhysRevLett.43.1754

A Single Snapshot Carries No Signal



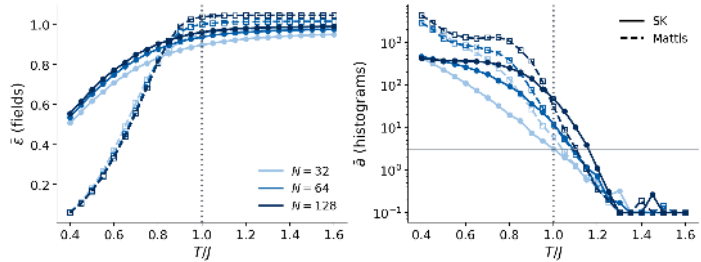
- $s_i \to \gamma_i s_i$, $J_{ij} \to \gamma_i \gamma_j J_{ij}$: pooled over disorder samples, a snapshot is a random string of signs
- left: largest PCA variance σ_1^2, one snapshot per sample (grey) or 64 per sample (orange); $N = 128$
- right: autoencoder error on new samples (solid) and training samples (dashed)

The Overlap Field Gives the Overlap



- PCA of q_i: uniform first direction, $w^{(1)} = \sqrt{N}q$, 11.0% of the variance ($N = 128$)
- autoencoder with one latent number, trained on $T \leq 0.6$: error $\approx 1 - q^2$ (dashed)
- steepest change at $T = 0.475, 0.425, 0.475$, far below T_c: no edge at the transition

The Mattis Model Isolates the Role of Frustration



- Mattis: $J_{ij} = \frac{J}{N}\zeta_i\zeta_j$, random signs $\zeta_i = \pm 1$, a hidden ferromagnet without frustration
- left: overlap-field autoencoder error (SK solid, Mattis dashed); right: score of a detector on the histogram of $|q|$, trained on $T \geq 1.3$, alarm level 3
- Mattis error at $T = 0.4$: 0.059 against 0.555 (SK, $N = 128$)

D. C. Mattis, Phys. Lett. A **56**, 421 (1976), doi:10.1016/0375-9601(76)90396-0